Applying Automotive EDR Data to Traffic Crash Reconstruction

TOYOTA Analysis



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Toyota

(includes Lexus and Scion)

- Toyota began phasing in EDR's in 2001 Lexus LS/GS – with precrash data
- Many early models have only longitudinal Delta V. By 2011 ALL have precrash data.
- Later models have 6 precrash data points, 5 at one second intervals, and a bonus point near TRG.
- In side impacts, some have side Delta V and in rollovers, some have roll angle

Toyota

(includes Lexus and Scion)

- Other manufacturers have all of their rules in the Data Limitations.
- Toyota is different. Some of the info found in other mfgr Data Limitations resides in the CDR Software's Glossary.
- For Toyotas, if you have a problem “RTFG” – Read the Friggin Glossary”

Toyota EDR Families

First Generation

- 00/02 with frontal crash only recording
- 00/02 with front and side/roll if equipped

Second Generation

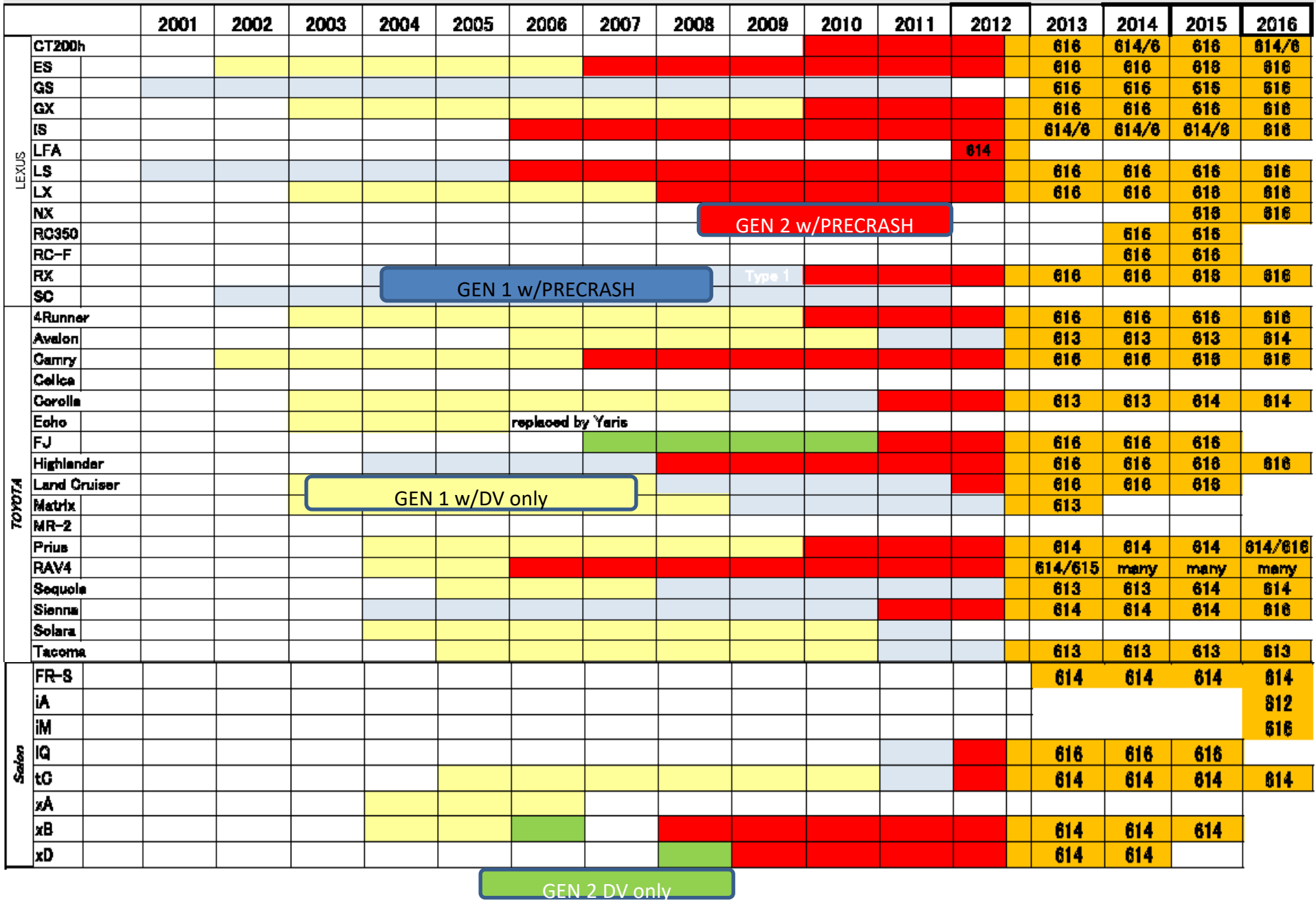
- 04/06 EDR

Third Generation (in US)

- 12 EDR = 563-intent.

Precrash data is not tied to a module family.

Toyota CDR 16.5 Coverage 74%



Toyota: Is event recording complete?

- Most pre-563-intent models will NOT make a new recording if power is lost at impact (same as GM/Ford/Chrysler).
- Exception: Delta V may be recorded real time as it is occurring.
- There is an Event Recording Complete data element for each major section of data.

Toyota: Is data from my crash?

- If power was lost at impact, probably not (until 2011+ MY)
- **Key Cycles** are NOT recorded until Part 563 Intent 2011+.
- You must use “rules for recording” to understand if data is from your event. Deployment data is locked and cannot be overwritten. Non-deployment events must meet a minimum 2G wakeup threshold, new ND events overwrite older ND events (see exception in Gen1).
- Gen 2 holds 2 frontal, 2 side, and 2 rollover events. Gen 1 holds 3 events, typically most recent 2 and most negative ND event experienced – see flow chart for exceptions.
- Events are NUMBERED – Event counter highest number is most recent event. Bosch CDR will report them in order (not so in older proprietary reports).
- Time between events up to register max value (5 sec for Gen 1, 16 sec for Gen 2)

From my event?
06 EDR Record Summary

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	7	Side Crash	0	Complete (Page 0)	Complete (Side Page 0)
1st Prior Event	6	Front/Rear Crash	-20	Complete (Page 0)	Complete (Front/Rear Page 1)
2nd Prior Event	5	Side Crash	-228	Complete (Page 0)	Complete (Side Page 1)
3rd Prior Event	4	Front/Rear Crash	-282	Complete (Page 0)	Complete (Front/Rear Page 0)

- Note front and side events often come in pairs if crash has an angular component to it. A less-than-default time between events indicates all events are part of the same sequence.

From my event?
02 EDR record summary

Front/Rear Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Event & Crash Pulse Data Recording Status
Most Recent Frontal/Rear Event	4	Front/Rear Crash	0	Complete (Front/Rear Page 1)
1st Prior Frontal/Rear Event	3	Front/Rear Crash	-5120 or greater	Complete (Front/Rear Page 2)
Prior Frontal/Rear Event	1	Front/Rear Crash	N/A	Complete (Front/Rear Page 0)

- 3 front/rear events not 2.
- 1st prior event is default value -5120ms or greater before most recent event, that means it is NOT RELATED – nor is #1, which is most likely the most negative event after most recent events #3 and #4

Pre-crash Data

- 6 Data Samples Precrash – 5 @1 sec intervals, 6th at TRG
- Sample before TRG to sample at TRG resolution is 0.1 sec.
- Data may be SAMPLED within 0.1 sec of TRG, but only refreshes EVERY 0.5 sec: **last point could still be up to 0.5 sec old.**
- Data Limitation: Speeds over **78 mph** (Gen 1) or 75 mph (Gen 2) are displayed as 78 or 75.

Pre-Crash Data

- Brake ON/OFF (same as all other mfrs)
- Accel Data NOT in % - in ***Voltage from sensor***
- Zero accel pedal is typically 0.78V, 100% is 3.79 V max seen to date but data limitations indicates the low and high could be different.
- You can calculate your own percentage if you want. %=(V-0.78)/(3.79-0.78)
- RPM in 400 RPM increments, 400-799=400, 800-1199=800, etc.

Precrash sampling rate

- If there are two events within 0.5 sec only one Pre Crash set of data is collected.
- Under rare circumstances, a precrash data set from a prior crash may be “orphaned” in memory and not be associated with any currently retained event.
- Speed data being sampled is refreshed every 500ms, accel pedal every 512ms, RPM every 24ms, Brake on as it occurs.

Delta V Data in Gen 2 models

- Longitudinal DV

  20 points @10ms intervals = 200ms duration starts at algorithm wake up - enough to get most crashes.
- Lateral DV (if equipped) for B pillar, C pillar and “Floor” (ACM) locations 25 points @4ms intervals – 100 ms duration starting 20ms before wakeup. Only use ACM (floor) location for recon.
- Roll Angle & Lateral Accel (if equipped).

  20 points @ 125 ms intervals beginning 625 ms before wake up

  Total duration 2.4 seconds
- Graph vs time, data table, and “Max” single value
- Same guidelines for use as GM/Ford

Other Useful Recon Data

- Front Driver and Pass Seat Belt Buckled
- Passenger Weight Classification (Empty/Unoccupied, Child, AF05 (small adult), AM 50 (midsize adult or larger)
- Driver Seat Position (FW forward vs RW)
- Gearshift lever position (P-R-N-Drive)
- What safety devices deployed when
- Time since prior event recorded (16381 ms max value default for Gen 2)

Toyota Gen 2 Front Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	YOUR VIN HERE
User	NONE
Case Number	NONE
EDR Data Imaging Date	
Crash Date	
Filename	52690 POLICE 38.ROT.CDRX
Saved on	
Collected with CDR version	Crash Data Retrieval Tool 4.0
Reported with CDR version	Crash Data Retrieval Tool 4.0
EDR Device Type	airbag control module
Event(s) recovered	Front/Rear (2)

Events: Front/Rear vs Side vs Rollover - & how many

Comments

Data imported from: 52690 POLICE 38.ROT

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items re
- Airbag ECU data should b
circumstances.
- If the airbags did not deplo called a Non-Deployment Event. D ecording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

P. 1 Same Look/Feel as others

Svstem Status at Retrieval

System Status at Time of Retrieval

ECU Part Number	89170-52690
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretenstioner Deployment
Diagnostic Codes Exist	No
Total number of Events	2
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent	2	Front/Rear Crash	0	Complete (Page 0)	Complete (Front/Rear Page 1)
1st Prior	1	Front/Rear Crash	-200	Complete (Page 0)	Complete (Front/Rear Page 0)

Back to back events

System Status at Event

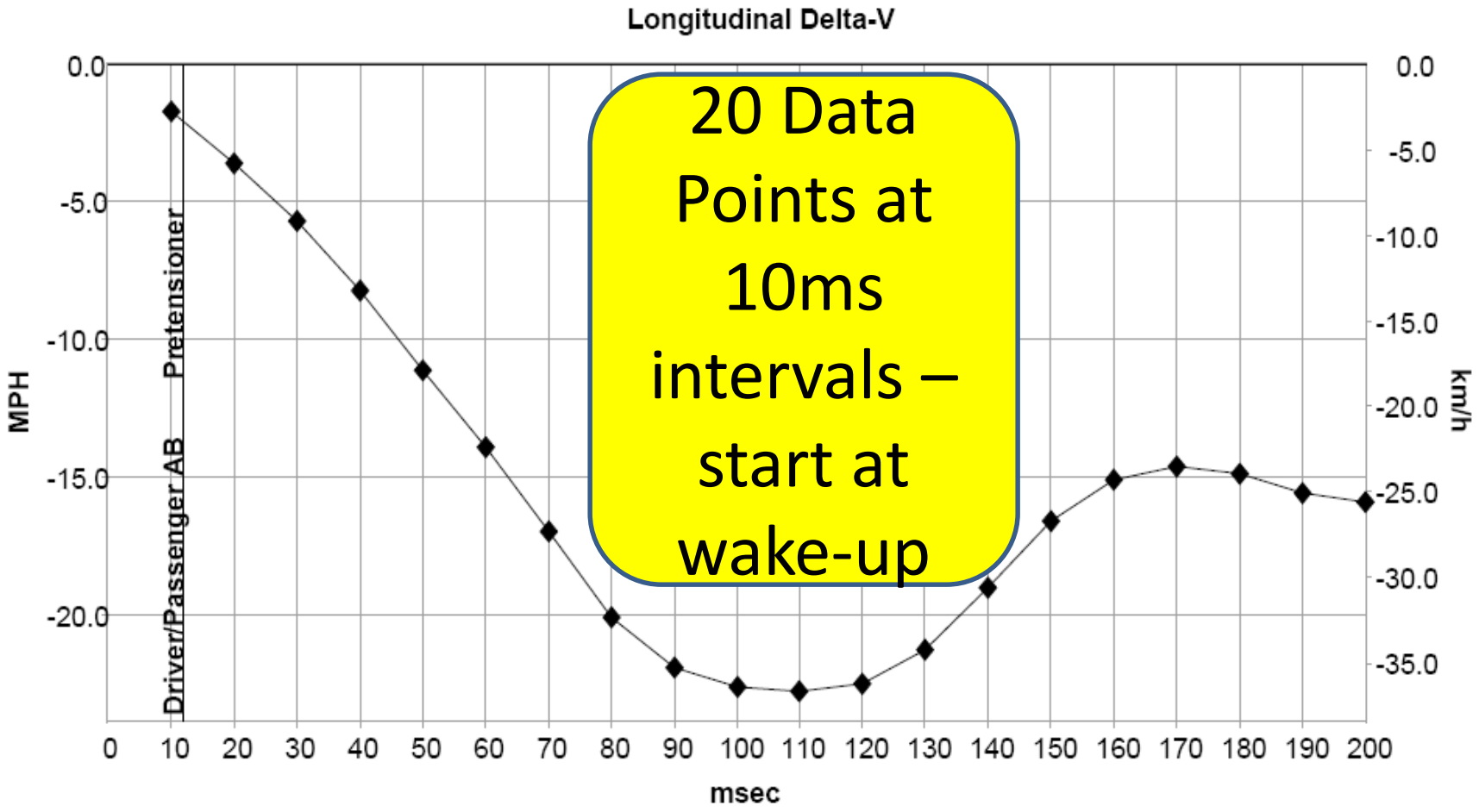
System Status at Event (Most Recent, TRG 2)

Recording Status, Front/Rear Crash Info.	Complete
Event Type	Front/Rear Crash
TRG Count (times)	2
Previous Crash Type	Frontal/Rear
Time from Pre-Crash TRG (msec)	200
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	12
Time to Deployment Command, Front Airbag, Passenger (msec)	12
Event Severity Status, Driver	Level 3
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	12

Delta V Data Graph

Longitudinal Crash Pulse (Most Recent, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH[km/h])	-22.8 [-36.7]



Delta V Data Table

Longitudinal Crash Pulse (Most Recent, TRG 2

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-1.7 [-2.7]
20	-3.6 [-5.8]
30	-5.7 [-9.2]
40	-8.2 [-13.2]
50	-11.1 [-17.9]
60	-13.9 [-22.4]
70	-17.0 [-27.4]
80	-20.1 [-32.3]
90	-21.9 [-35.2]
100	-22.6 [-36.4]
110	-22.8 [-36.7]
120	-22.5 [-36.2]
130	-21.3 [-34.3]
140	-19.0 [-30.6]
150	-16.6 [-26.7]
160	-15.1 [-24.3]
170	-14.6 [-23.5]
180	-14.9 [-24.0]
190	-15.6 [-25.1]
200	-15.9 [-25.6]

20 Data Points at 10ms intervals – start at wake-up

Pre Crash Data - Overview

DTCs Present at Time of Event (Most Recent, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	200
Buckle Switch, Driver	Unbuckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Pre Crash Data Timing

Other Points a 1 sec intervals

6th point Sampled at TRG

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

Pre Crash Data - Overview

DTCs Present at Time of Event (Most Recent, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

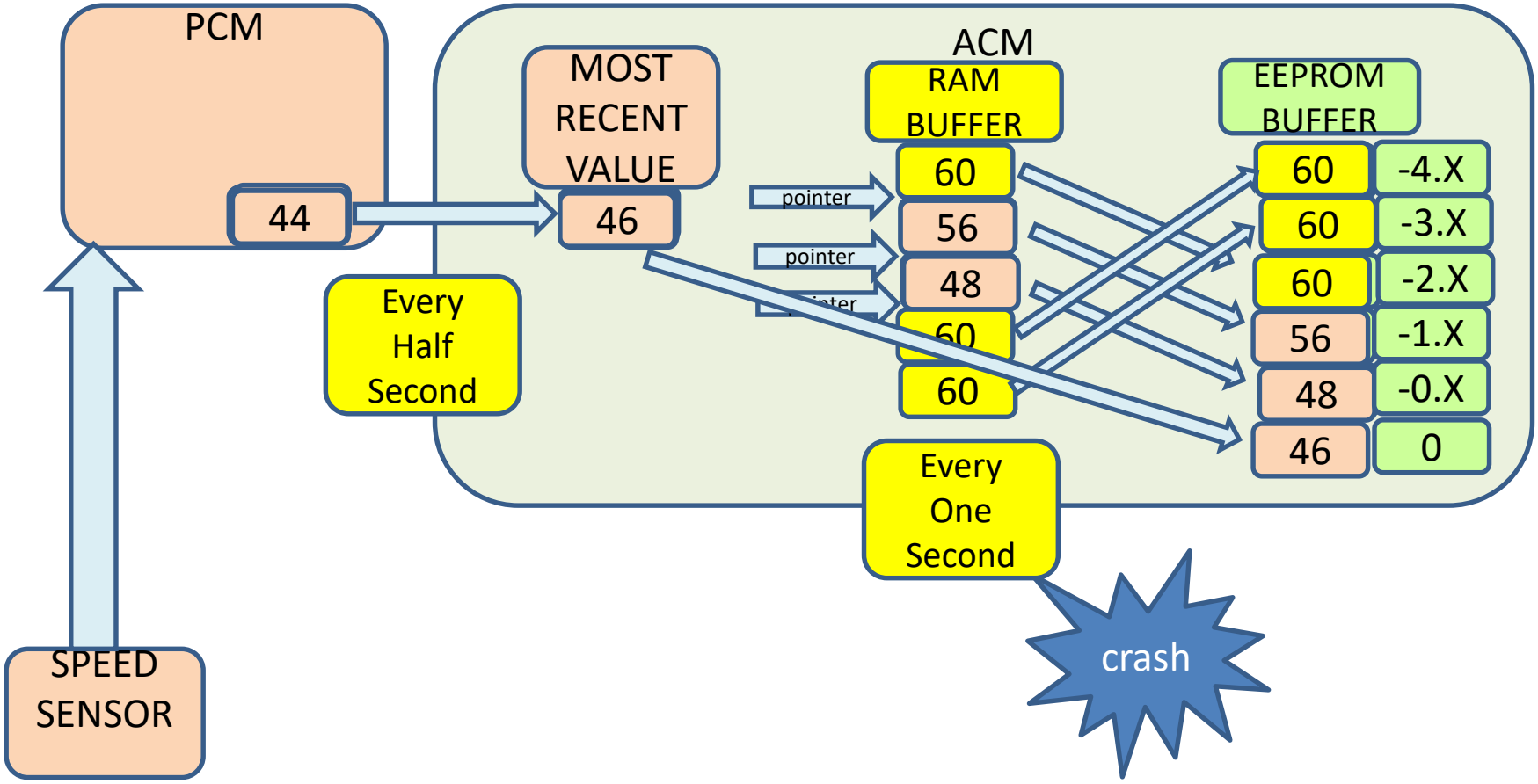
Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	200
Buckle Switch, Driver	Unbuckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Toyota speed data transfer[34]



Pre Crash Data Timing

Other Points a 1 sec intervals

6th point Sampled at TRG

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

Pre Crash Speed Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Since the speed data buffer is only refreshed every 0.5 sec, the 0 and -0.2 data points may be from the same buffer value. If the time between TRG and the prior point is less than 0.5 sec and the values are identical, especially during precrash braking, do not assume the vehicle has stopped slowing down.
- Resolution 2km/hr (1.2 mph)
- Max Value 75 mph (121kph) on Gen 2 (04/06), (78.3mph in Gen 1 00/02 models)

Pre Crash Accel Pedal Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Only refreshed every 512 ms – held in buffer
- Volts not pedal percent
- Convert to % = (V-0.78)/(3.79-0.78)
- Data Limitations warns min & max may vary

Pre Crash RPM Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Resolution to next lower 400 rpm
- If speed is reported at max 75 mph, can be used to calculate a higher min speed based on gear ratios.
- Speed in mph = RPM/trans ratio/final drive ratio/tire revs /mile*60

- Next Event

System Status at Event (1st Prior, TRG 1)

Recording Status, Front/Rear Crash Info.	Complete
Event Type	Front/Rear Crash
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	Not Commanded
Time to Deployment Command, Front Airbag, Passenger (msec)	Not Commanded
Event Severity Status, Driver	N/A
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	Not Commanded

RTFDL Caution

- In the 00 EDR, seat belt status in ND events is the status at time of READOUT, not the time of EVENT.
- This is duly noted in the data limitations, but has tripped some people up. A module was read in car that said unbelted, then re-read on the benchtop that gave the default value of belted.

- For Non-Deployment Event, "Buckle Switch", "Ignition Cycle Since DTC was Set" and "Airbag Warning Lamp ON Time Since DTC was Set" is the ACM parameter data at the time of retrieval, not the parameter data when the recording trigger was established. For Deployment Event, it is the information when the airbag deployment.

Toyota Gen 2 Side Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

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CDR File Information

User Entered VIN	4T1BF3EK4BU664098
User	Richard R. Ruth, P.E.
Case Number	Gazdick salvage yard find
EDR Data Imaging Date	06/01/2011
Crash Date	
Filename	4T1BF3EK4BU664098_ACM_GAZDICK_BODY_SHOP.CDRX
Saved on	Wednesday, June 1 2011 at 21:52:48
Collected with CDR version	Crash Data Retrieval Tool 4.0
Reported with CDR version	Crash Data Retrieval Tool 4.0
EDR Device Type	airbag control module
Event(s) recovered	Front/Rear (1), Side (1)

Comments

imaged direct to module with cable 614 and 387 adaptor

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

Events: Front/Rear vs Side vs Rollover - & how many

System Status **at Side Event**

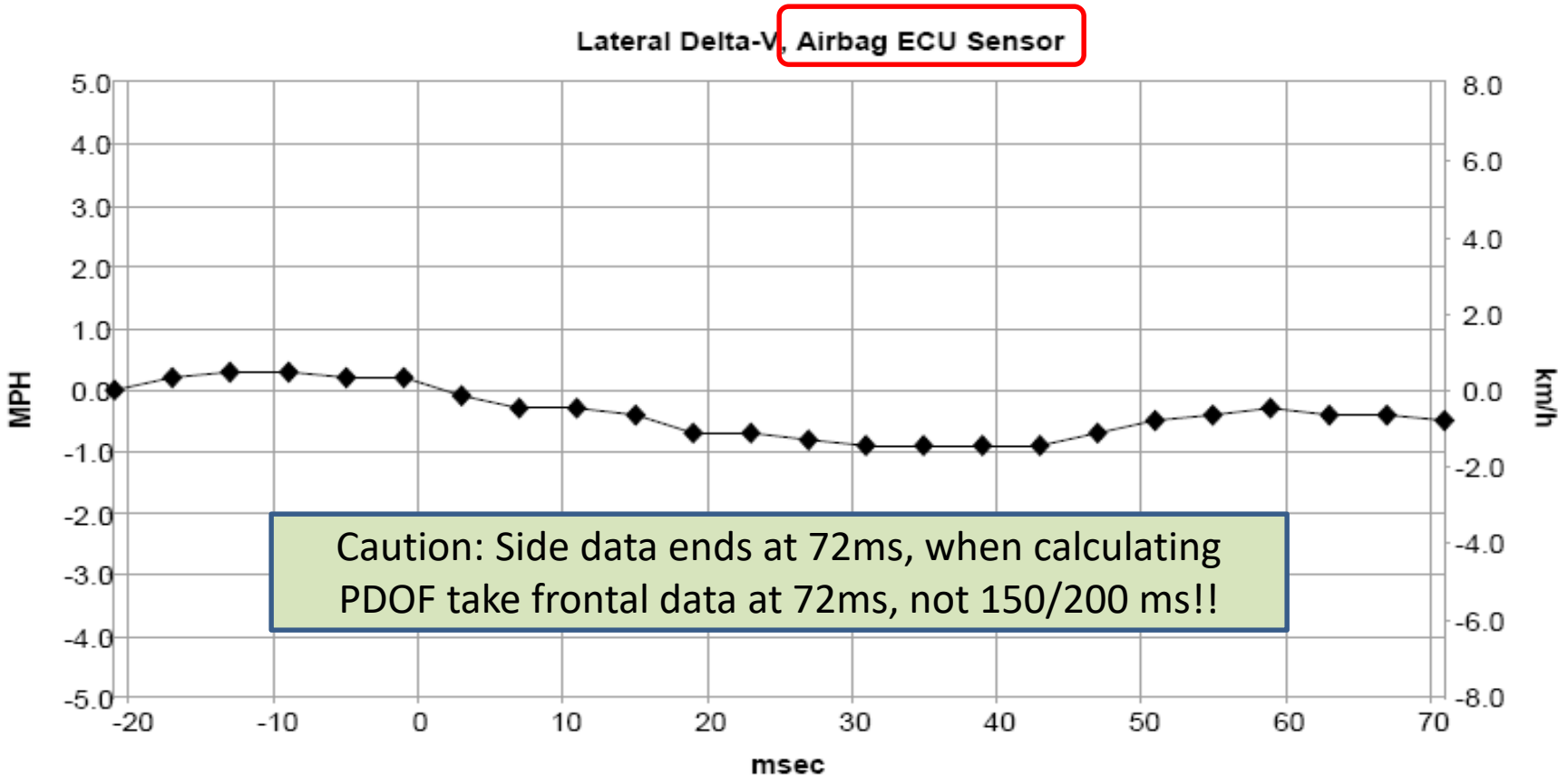
System Status at Event (Most Recent, TRG 2

Recording Status, Side Crash Info.	Complete
Event Type	Side Crash
TRG Count (times)	2
Recorded Side	Passenger's Side
Previous Crash type	Frontal/Rear
Time from Pre-Crash TRG (msec)	24
Linked Pre-Crash Page	0
Time to Deployment Command, Row 1 (msec)	Not Commanded
Time to Deployment Command, Row 2 (msec)	Not Commanded

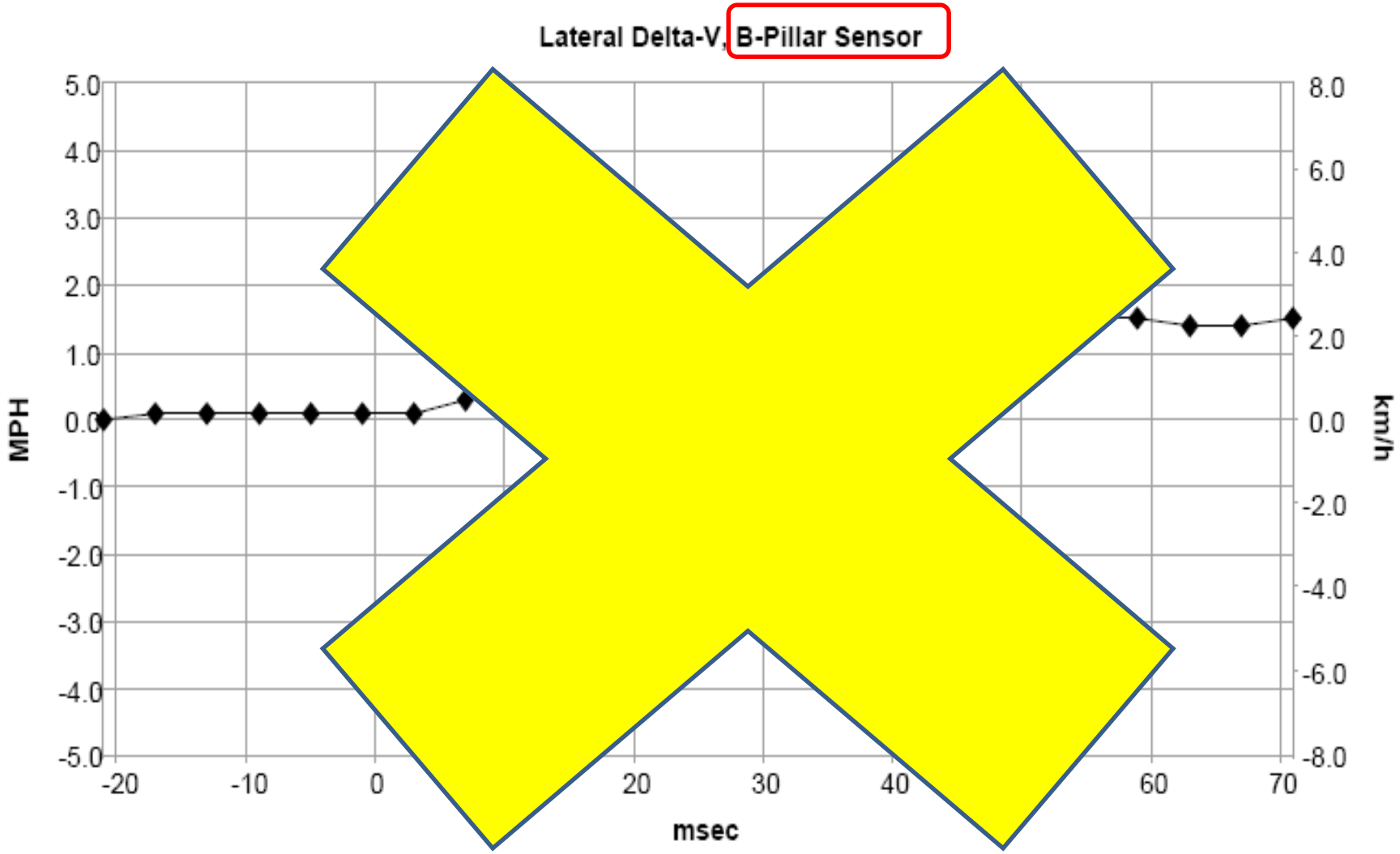
Lateral Delta V Graph of ECU Sensor

Lateral Crash Pulse (Most Recent, TRG 2 - table 1 of 2)

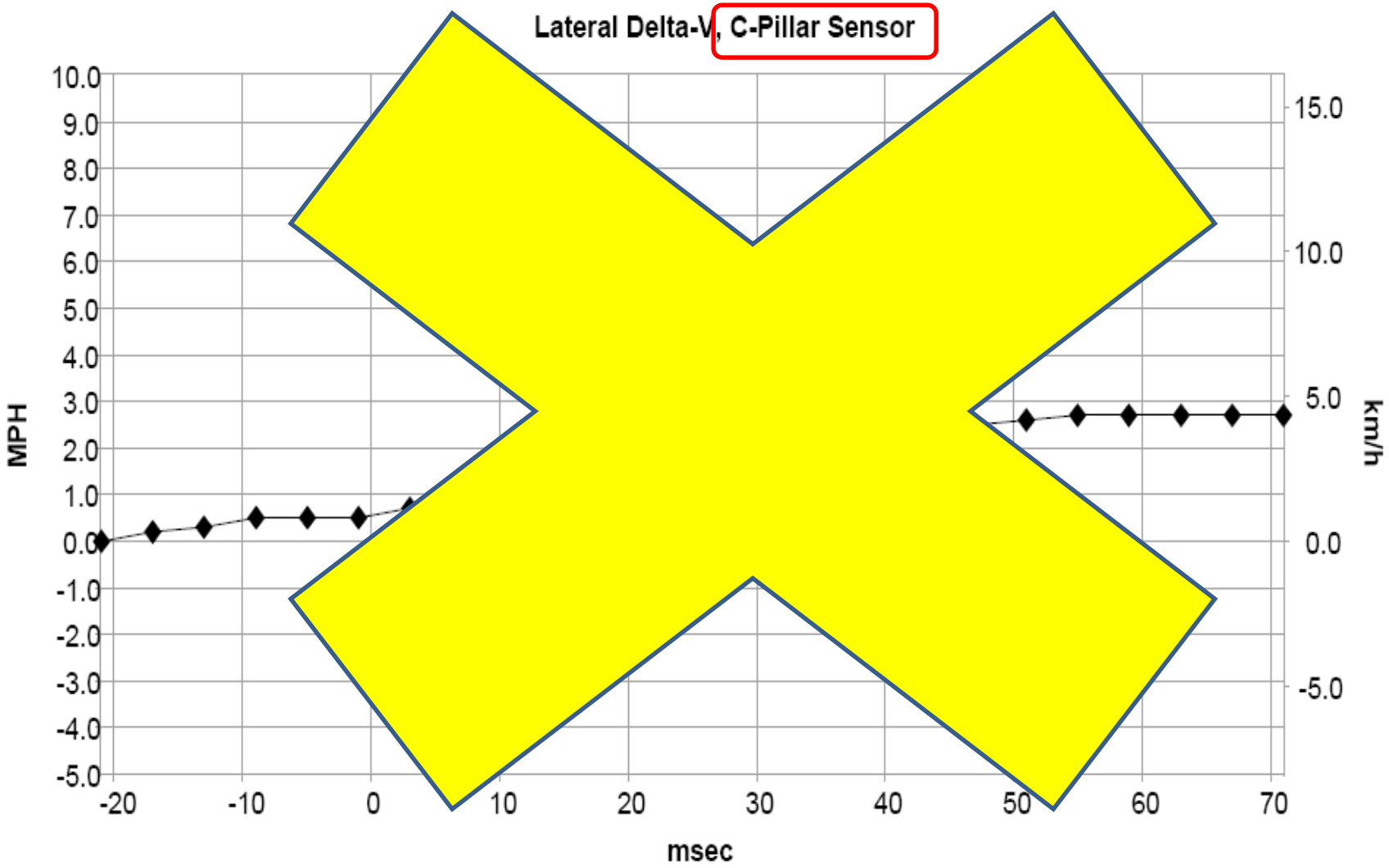
Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	3
Max Lateral Delta-V, Floor (MPH[km/h])	-0.9 [-1.4]
Max Lateral Delta-V, B-Pillar Sensor (MPH[km/h])	1.5 [2.4]
Max Lateral Delta-V, C-Pillar Sensor (MPH[km/h])	2.7 [4.3]



Don’t Use B Pillar Side Data



Don’t Use C Pillar Side Data



Lateral Crash Pulse (Most Recent, TRG 2 - table 2 of 2)

Time (msec)	Lateral Delta-V, Airbag ECU Sensor (MPH [km/h])	Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	Lateral Delta-V, C-Pillar Sensor (MPH [km/h])
-21	0.0 [0.0]	0.0 [0.0]	0.0 [0.0]
-17	0.2 [0.3]	0.1 [0.2]	.2 [0.3]
-13	0.3 [0.5]	0.1 [0.2	3 [0.5]
-9	0.3 [0.5]	0.1 [0	[0.8]
-5	0.2 [0.3]	1 [	[0.8
-1	0.2 [0.3]	1	0
3	-0.1 [-0.2]		
7	-0.3 [-0.5]		
11	-0.3 [-0.5]		
15	-0.4 [-0.6]		
19	-0.7 [-1.1]		1
23	-0.7 [-1.1]		
27	-0.8 [-1.3]		
31	-0.9 [-1.4]		
35	-0.9 [-1.4]		
39	-0.9 [-1.4]	3	3.
43	-0.9 [-1.4]	.3 [2	[3.9
47	-0.7 [-1.1]	1.5 [2.	[4.0]
51	-0.5 [-0.8]	1.5 [2.4	.6 [4.2]
55	-0.4 [-0.6]	1.5 [2.4]	2.7 [4.3]
59	-0.3 [-0.5]	1.5 [2.4]	2.7 [4.3]
63	-0.4 [-0.6]	1.4 [2.3]	2.7 [4.3]
67	-0.4 [-0.6]	1.4 [2.3]	2.7 [4.3]
71	-0.5 [-0.8]	1.5 [2.4]	2.7 [4.3]

Pre Crash Data Page (looks same for side and frontal)

DTCs Present at Time of Event (Most Recent, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	700
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.7	-3.7	-2.7	-1.7	-0.7	0 (TRG)
Vehicle Speed (MPH [km/h])	46 [74]	47.2 [76]	47.2 [76]	47.2 [76]	42.3 [68]	24.9 [40]
Brake Switch	OFF	OFF	OFF	OFF	ON	ON
Accelerator Rate (V)	1.25	1.09	0.98	0.98	0.78	0.78
Engine RPM (RPM)	1,600	1,200	1,200	1,200	800	800

Toyota Speed at Impact

TWO changes from 42 to 24?

PRE-CRASH DATA						
Time	-4.7	-3.7	-2.7	-1.7	-0.7	0
Speed	46 (74)	47.2 (76)	47.2 (76)	47.2 (76)	42.3 (68)	24.9 (40)
Brake	Off	Off	Off	Off	On	On
Time Range	-5.2 to -4.7	-4.2 to -3.7	-3.2 to -2.7	-2.2 to -1.7	-1.2 to -0.7	0.2 to 0.0

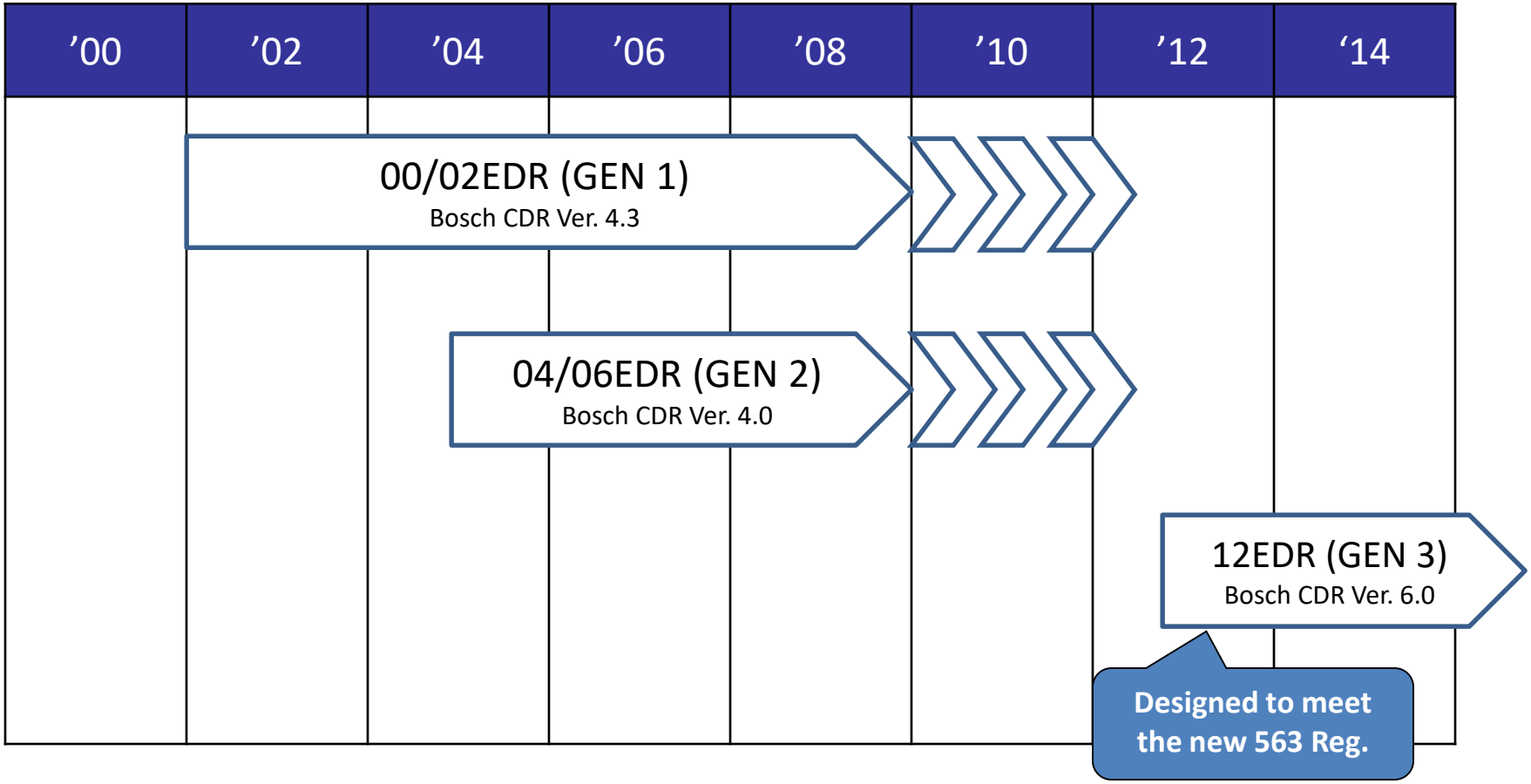
SPEED AT IMPACT WORKSHEET		English		Metric	
		MIN	MAX	MIN	MAX
Last Good Data Point mph (km/h)		24.9	24.9	40	40
Braking	-0.2 sec* 15 mph/sec	-3.0	0	-4.8	0
Slip +5%		1.2	1.2	2	2
Speedo +/-4%		-1	1	-1.6	1.6
Range		22.1	27.1	35.6	43.6

Age of TRG data sample

Time Last Point to TRG	Changed twice	Changed once	Didn't change
0	n/a	0	0.5
0.1	n/a	0.1	0.5
0.2	n/a	0.2	0.5
0.3	n/a	0.3	0.5
0.4	n/a	0.4	0.5
0.5	0	0.5	n/a
0.6	0.1	0.5	n/a
0.7	0.2	0.5	n/a
0.8	0.3	0.5	n/a
0.9	0.4	0.5	n/a

- Data at TRG can be 0.5 sec old, but may be LESS than 0.5 sec if data has changed ONCE from prior sample within 0.5 sec
- Data at TRG can ALSO be LESS than 0.5 sec if data has changed TWICE from prior sample.
- Version 2.01 of the speed at impact worksheet excel program does this for you

History of Toyota EDR

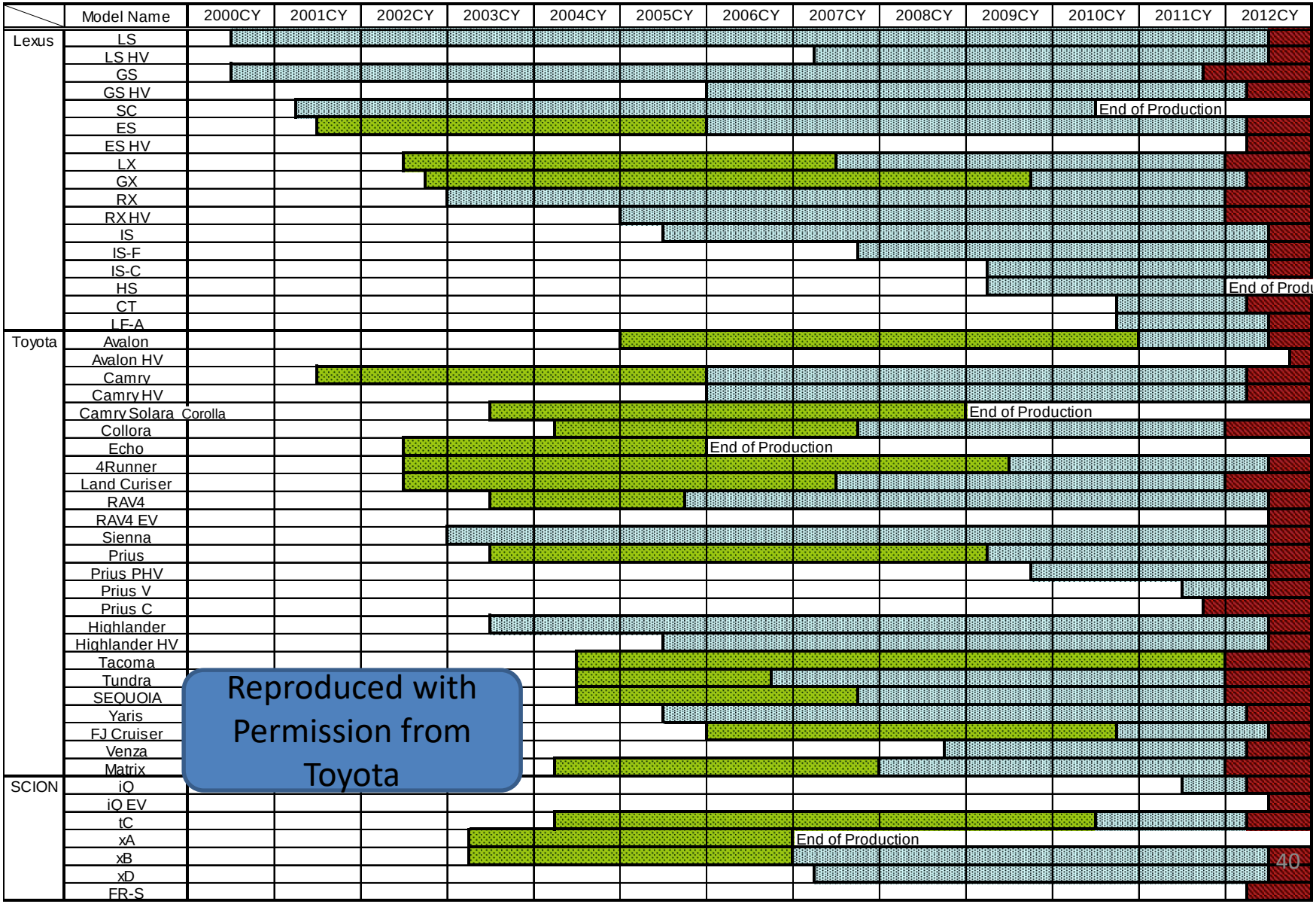


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Reference the CDR Help Menu for specific vehicle EDR information

EDR Installation (North America Only)

: **Post-crash data only** : **Pre- & Post-crash data** : **Part563 EDR**



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Gen 3 -Part 563 Intent Toyota”12EDR”

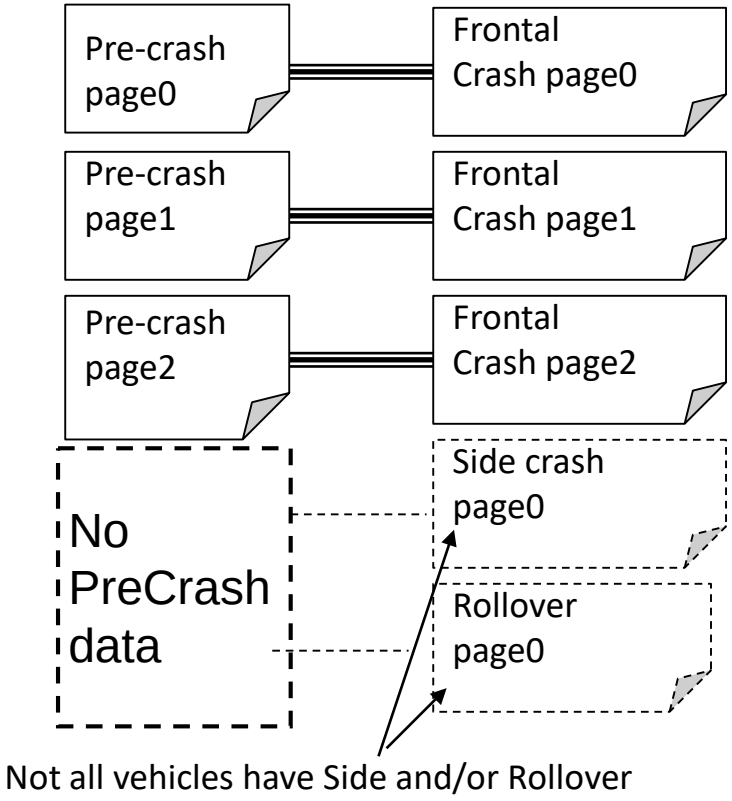
- Phase in mid 2012 MY
- Adds key cycles, deploy times, longer ΔV (250ms), more pre-crash 5 sec @ 0.5 sec intervals, speed to 122 mph (200 KPH), Accel %.
- Lat ΔV only recorded in side events (Unique to Toyota)
- Recording threshold 5mph ΔV min.
- Adds Throttle %, Steering, Yaw Rate, Brake Pressure, VSC Long Accel, Cruise, Gear, RPM @ 100 rpm resolution.

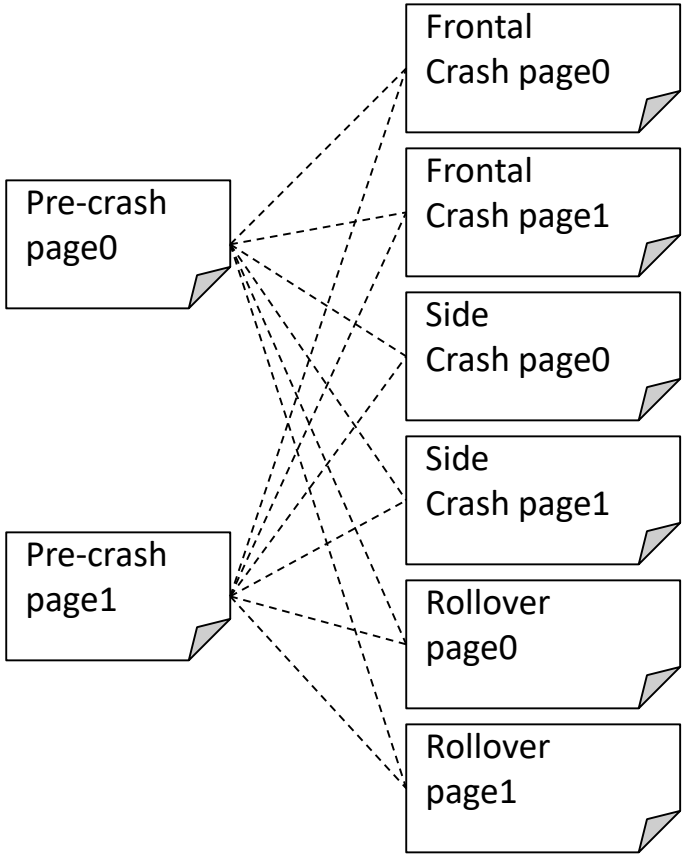
Gen 3 “12 EDR”

- Speed data updated more frequently now - every 20 to 24 ms – (64 ms on accel pedal)
- Toyota continues to sample data point at TRG
- Brake **pressure** data element added in MPa
- Accel pedal data comeS in % vs. volts
- Analyze the same as any other 563 intent module, except recognize that last data point is for all practical purposes at TRG now.

12 EDR "Pages"

00/02EDR (GEN 1)	04/06EDR (GEN 2) and **12EDR (GEN 3)**
▪Frontal Crash: **3** pages ▪Side and Rollover: 1page (no Pre-Crash)	▪Pre-Crash : **2** pages ▪Frontal Side Rollover :2 pages each





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Pre-Crash Parameter Comparison

Parameter	04/06 EDR		12 EDR	
Vehicle Speed (Part 563)	2 km/h /LSB	0~122 [km/h]	1 km/h /LSB	0 ~ 200 [km/h]
Accelerator Pedal (Part 563)	.039 V/LSB	0~5 [V]	0.5 %/LSB	0 ~ 100 [%]
Engine RPM (Part 563)	400 RPM/LSB	0~5200 [RPM]	100 RPM/LSB	0 ~ 12800 [RPM]
Service Brake (Part 563)	ON/OFF		ON/OFF	
Shift Position	R / N/ P / D		R / N / P / D / 5 / 4/ 3 / 2/ LO/ B/ Seq	
Percentage of Engine Throttle			0.5%/LSB	0 ~ 100 [%]
Motor RPM (Part 563)			100 RPM/LSB	-7500 ~ 17500 [RPM]
Brake Oil Pressure			0.048 MPa	0 ~ 12.144 [MPa]
Sequential Shift			1 / 2 / 3 /4 / 5 / 6 / 7 / 8	
Steering Input (Part 563)			3 deg/LSB	-375 ~ 375 [deg]
VSC Yaw Rate			0.488 deg/s/LSB	-61 ~ 61 [deg/s]
VSC Longitudinal Acceleration			0.07178 m/s^2/LSB	-8.9725~8.9725 [m/s^2]
Cruise Control Status			ON/OFF	
Drive Mode - Power Drive Mode – Eco Drive Mode - Sport Drive Mode - Snow Drive Mode - EV			ON/OFF	

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Gen 3 Longitudinal Stability Control Accel Data

- Units are in m/sec^2, divide by 9.8 to get G's (dividing by 10 gets you close)
- Data is like a Vericom, similar to Ford stability control system data, just not as frequent.
- It is not yet clear if the accel data is instantaneous or average over the interval, but if multiple consecutive data points are available averaging is recommended

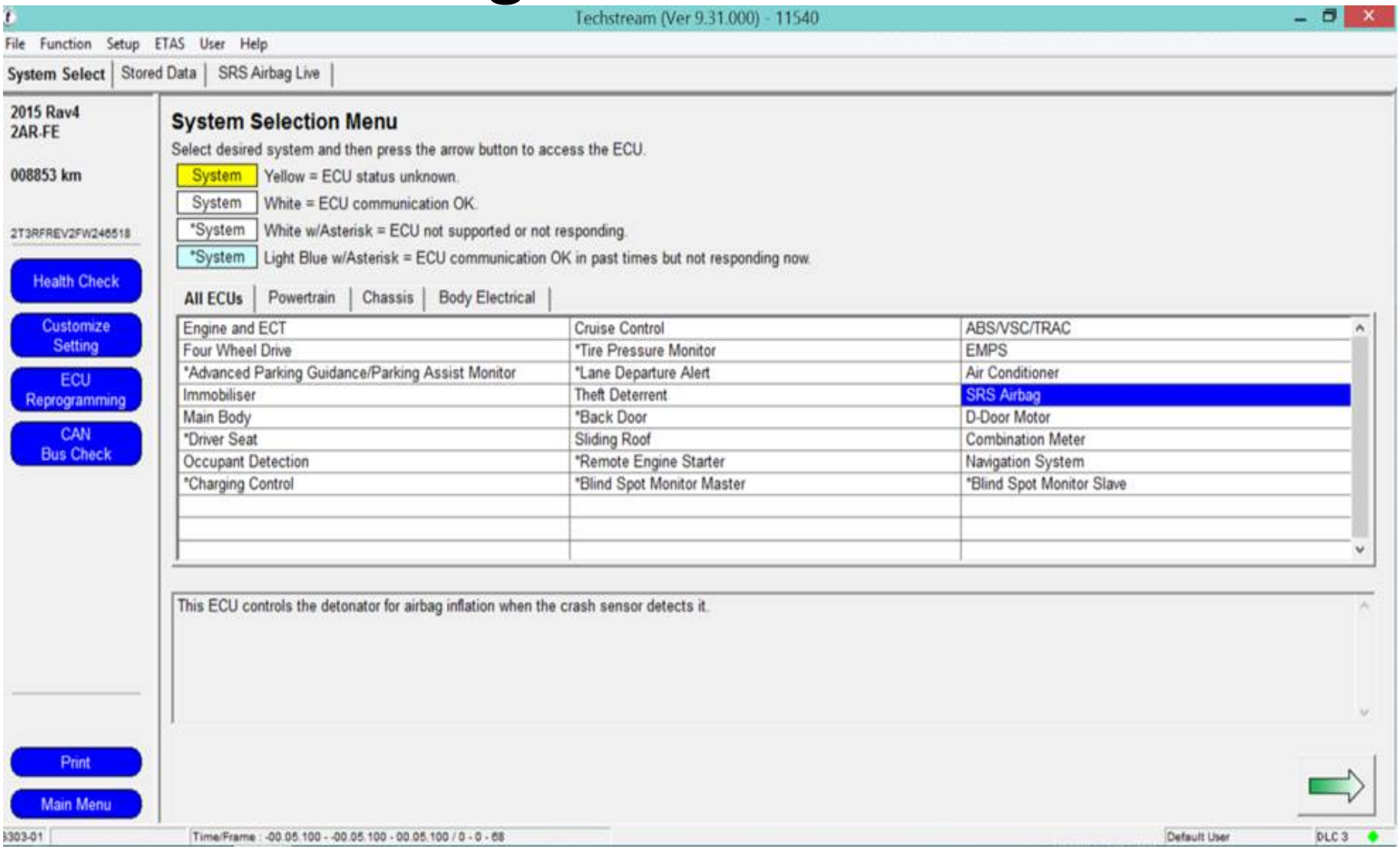
New “13 EDR” & “15 EDR”

- 13 EDR Adds lateral Delta V data from ACM (“floor sensor”) in frontal crashes (Fixes biggest drawback to 12 EDR), and lateral accel from ACM in side crashes.
- Lateral acceleration data table is in meters/sec^2, you may need to calculate Delta V from it yourself.
- Remember side data duration of recording is LESS than frontal direction –you may not be getting the WHOLE crash
- 15 EDR adds side impact door pressure sensor models

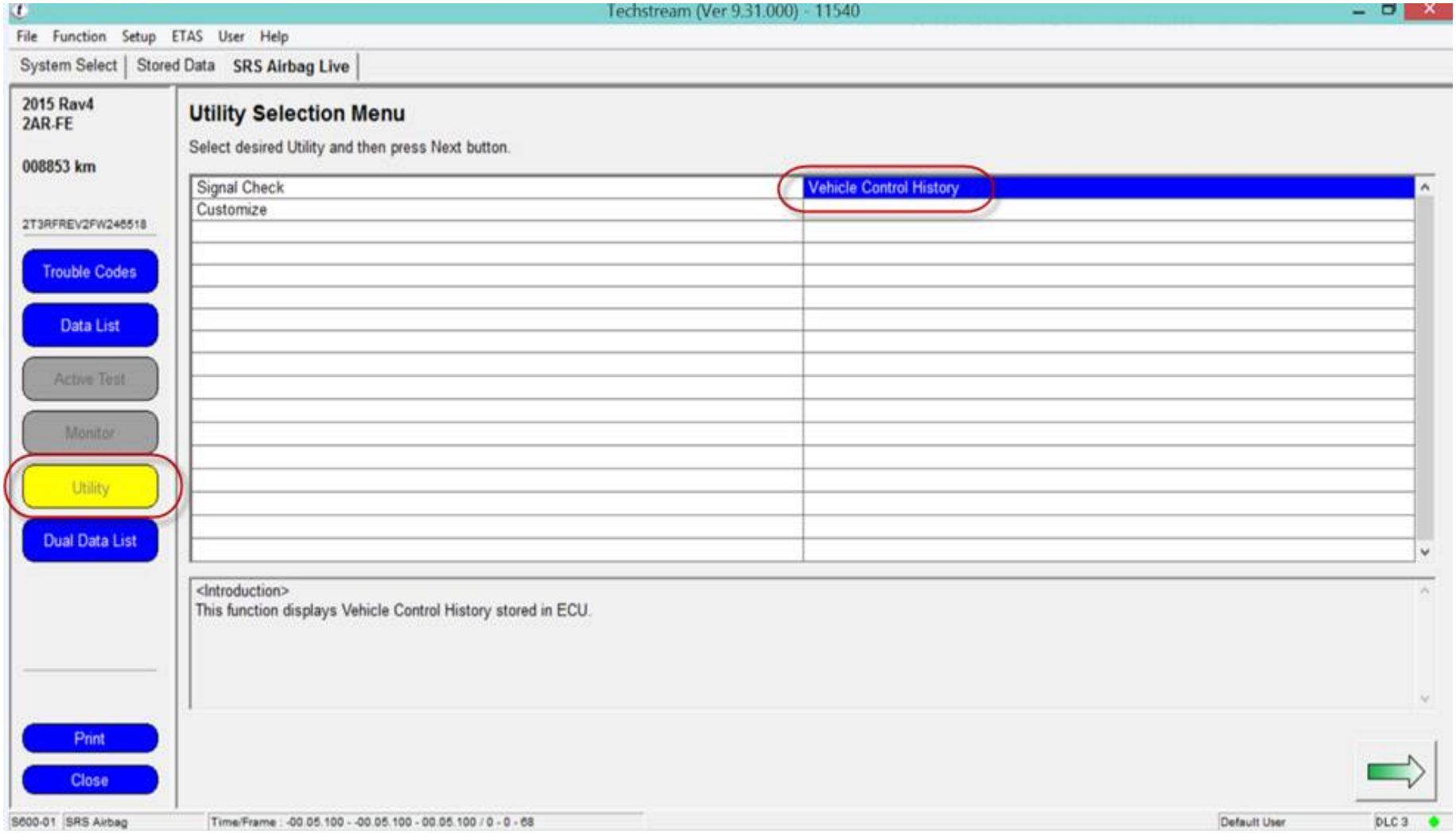
New Freeze Frame Data

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, “Vehicle Control History”
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a history of erratic driver behavior
- Access this whenever EDR data is either not available or insufficient for your investigation

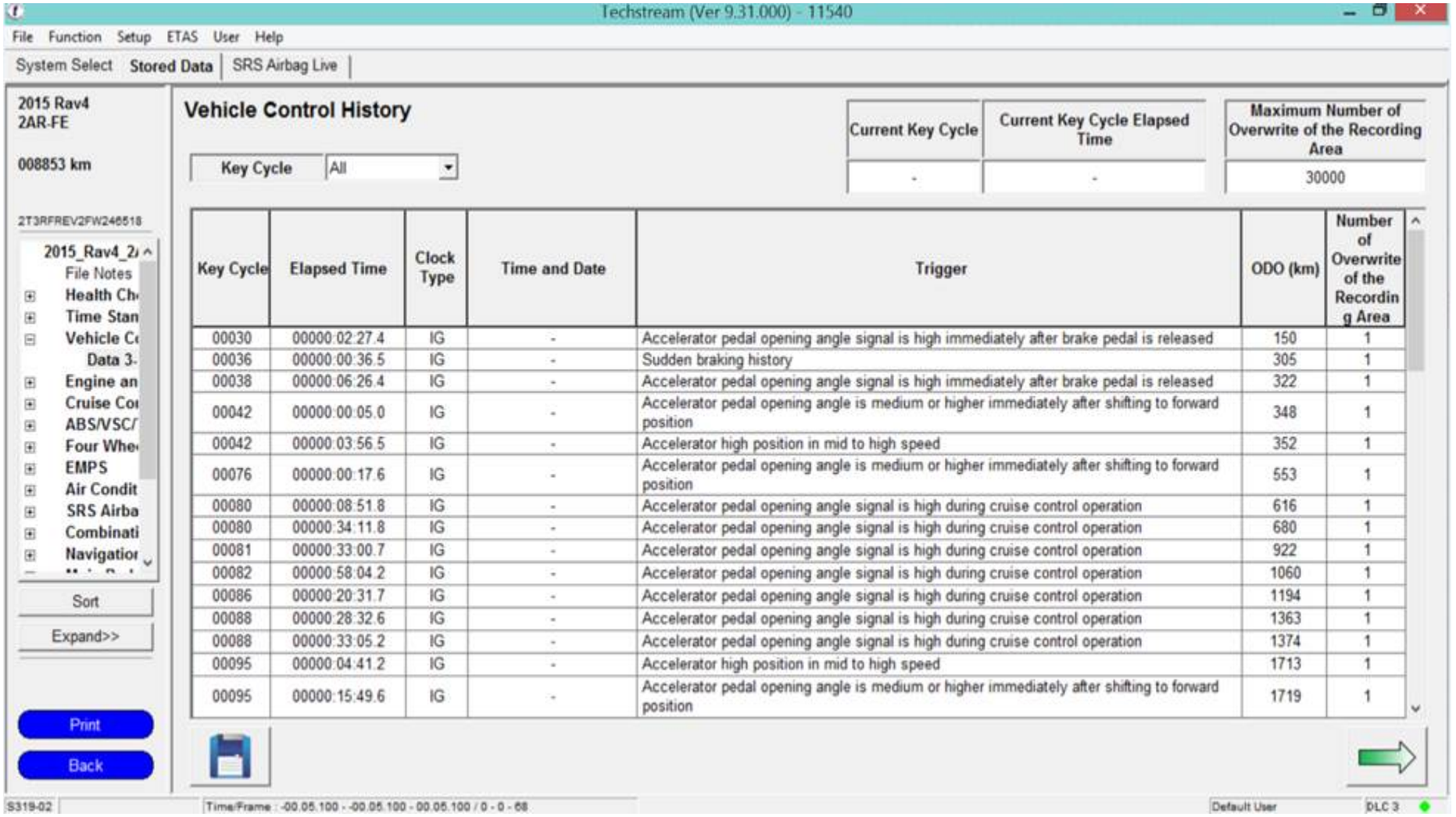
Getting to Vehicle Control History Using Techstream Tool



Using Techstream to Access VCH data

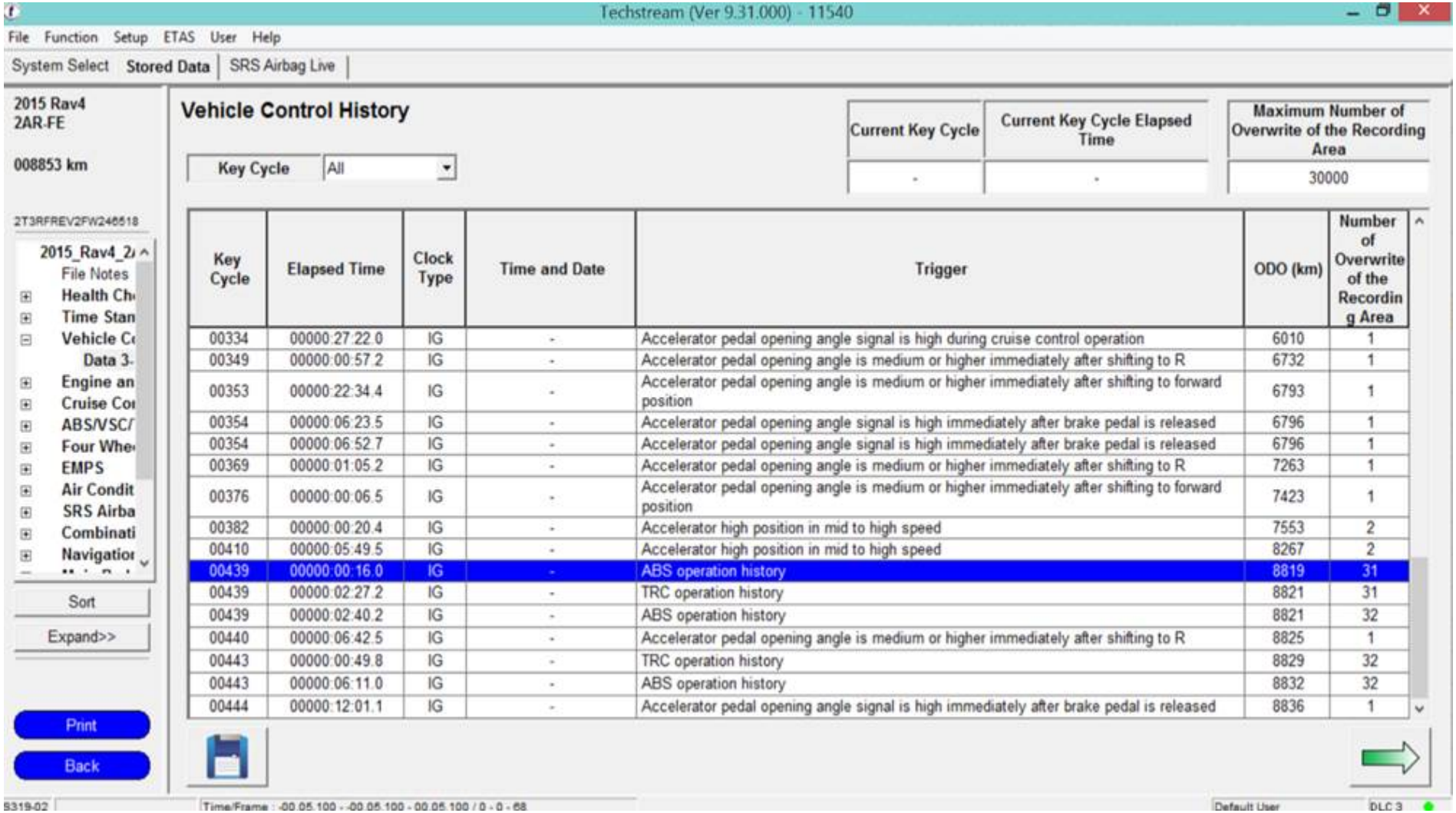


VCH List of Recorded Events



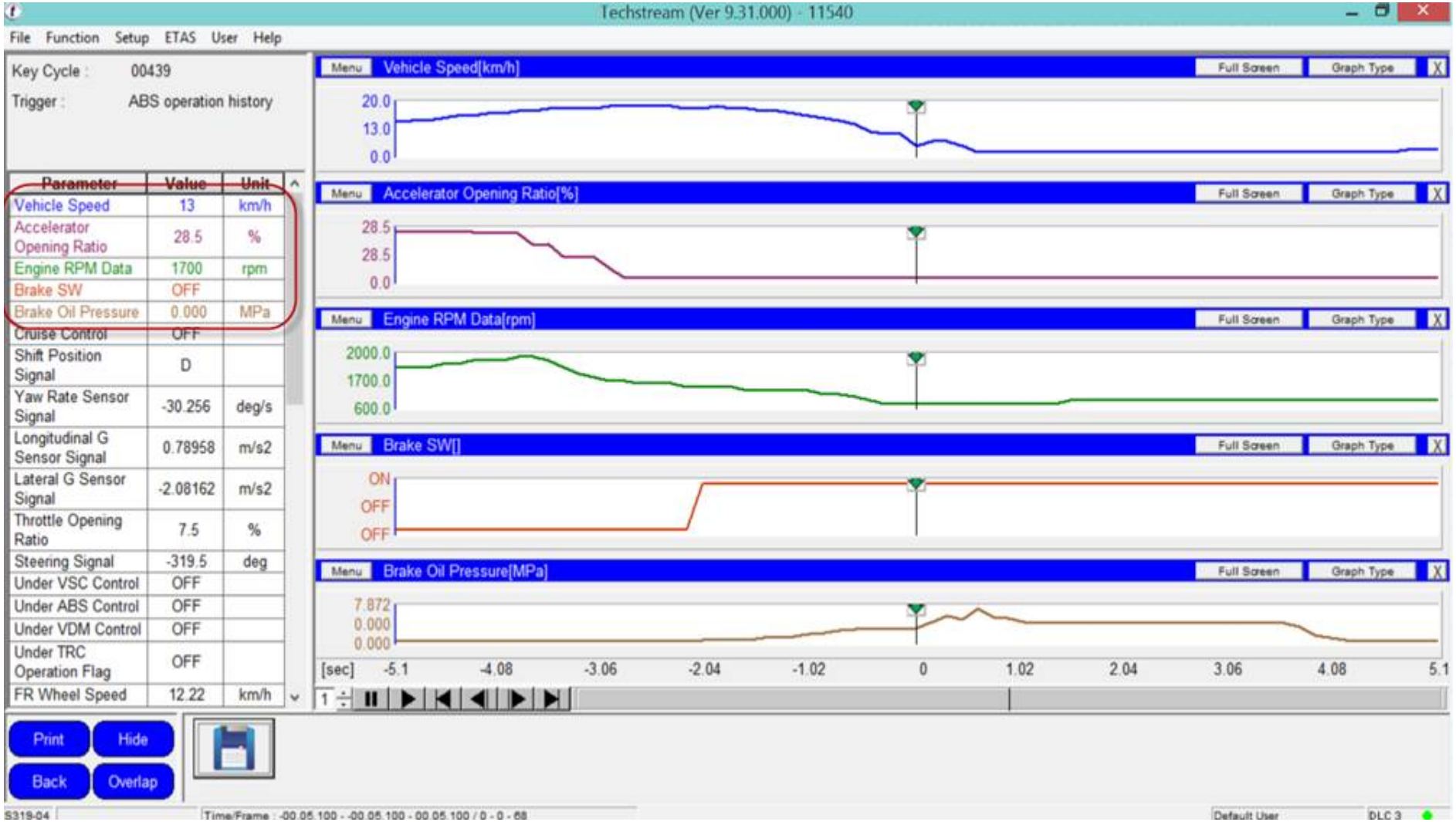
Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (km)	Number of Overwrite of the Recording Area
00030	00000:02:27.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	150	1
00036	00000:00:36.5	IG	-	Sudden braking history	305	1
00038	00000:06:26.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	322	1
00042	00000:00:05.0	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	348	1
00042	00000:03:56.5	IG	-	Accelerator high position in mid to high speed	352	1
00076	00000:00:17.6	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	553	1
00080	00000:08:51.8	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	616	1
00080	00000:34:11.8	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	680	1
00081	00000:33:00.7	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	922	1
00082	00000:58:04.2	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1060	1
00086	00000:20:31.7	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1194	1
00088	00000:28:32.6	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1363	1
00088	00000:33:05.2	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1374	1
00095	00000:04:41.2	IG	-	Accelerator high position in mid to high speed	1713	1
00095	00000:15:49.6	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	1719	1

Highlight Entry To Get Detail



Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (km)	Number of Overwriting Area
00334	00000:27:22.0	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	6010	1
00349	00000:00:57.2	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	6732	1
00353	00000:22:34.4	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	6793	1
00354	00000:06:23.5	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	6796	1
00354	00000:06:52.7	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	6796	1
00369	00000:01:05.2	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	7263	1
00376	00000:00:06.5	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	7423	1
00382	00000:00:20.4	IG	-	Accelerator high position in mid to high speed	7553	2
00410	00000:05:49.5	IG	-	Accelerator high position in mid to high speed	8267	2
00439	00000:00:16.0	IG	-	ABS operation history	8819	31
00439	00000:02:27.2	IG	-	TRC operation history	8821	31
00439	00000:02:40.2	IG	-	ABS operation history	8821	32
00440	00000:06:42.5	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	8825	1
00443	00000:00:49.8	IG	-	TRC operation history	8829	32
00443	00000:06:11.0	IG	-	ABS operation history	8832	32
00444	00000:12:01.1	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	8836	1

Detailed VCH Data Per Event



Case Study Using Toyota Data

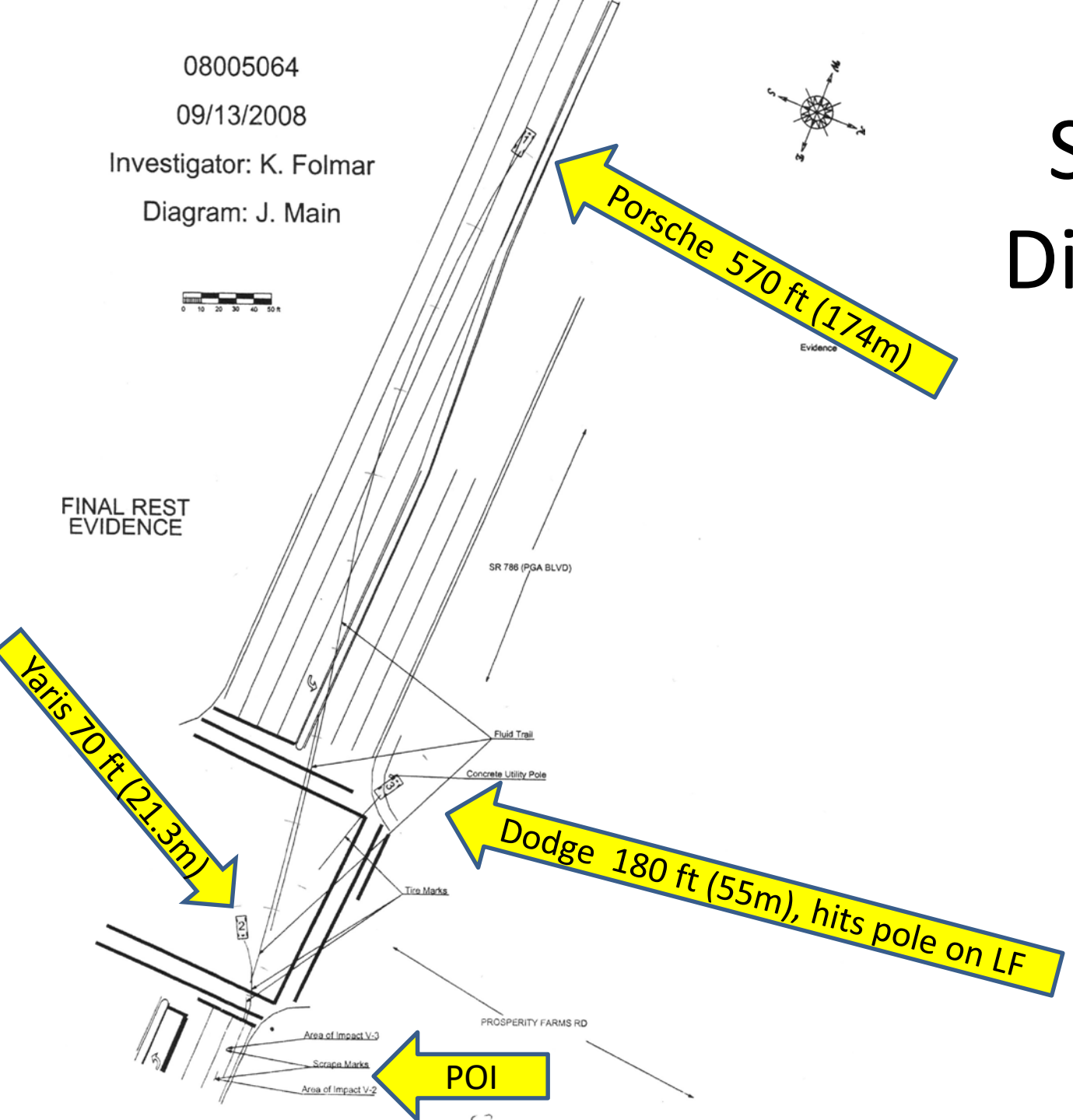


- 2 cars are Stopping/Stopped at an intersection
- 3rd Car in, Porsche Cayenne, does not stop
- Porche hits back of 2008 Yaris – mostly inline
- Yaris gets pushed into back of Dodge
- Yaris driver “sandwiched”, fatal
- Reconstruction Issues: How fast was Porsche going at impact?

Teaching Points of This Case

- The best data recorder may not be in the defendant's car, but in the car struck.
- For an inline crash, Longitudinal Delta V and vehicle weights give you closing speed.
- With Closing Speed and Speed of car struck, you can get approach speed of defendant.
- Toyota speed at impact is more precise than many other manufacturers.

Scene Diagram



Porsche into Yaris



Yaris into Dodge



Defendant's Porsche Front Damage



2008 Toyota Yaris Damage



Yaris Rear Damage



Yaris Front Damage



Dodge Damage



Dodge mounts curb, hits pole



Slide to Stop and Momentum

Slide to stop speeds for the 3 vehicles based on POI/POR

- 95 Intrepid 180 ft, f=.7, 25%+ brake & pole energy equivalent = 32mph (51.5 kph)
- 08 Yaris goes 70 ft, f=.7, 100% brake = 38 mph (61 kph)
- 06 Cayenne goes 580 ft, f=.7, 12% brake = 38 mph (61)
- Common exit velocity of all 3 avg's 36 mph (58 kph)

Momentum

- Intrepid 4472 lbs, Yaris 2286 lbs, Cayenne 5687 lbs
- (Intrepid 2030 Kg, Yaris 1038 Kg, Cayenne 2581 Kg)
- Porsche weight * speed = Weight of all 3 * exit velocity
- 5687*X = (5687 +2286 + 4472)*36 mph
- X = (12445/5687)*36 = 79 mph (127 kph) speed of Porsche

Event Data Recorder Analysis

- The Defendant's vehicle, an 06 Porsche Cayenne, did not have any accessible data recorder
- The first car the defendant hit, the 2008 Toyota Yaris, was not supported by the Bosch Crash Data Retrieval System
- The investigating officer contacted Toyota, who agreed to assist Law Enforcement
- A report was received but ***there were no data limitations with it***.

Was the Recording from My Crash?

- This Toyota EDR did **NOT** have key cycles at event or investigation – and does require power after a crash to complete a recording.
- Vehicle electrical appeared intact post crash.
- Two events captured – a severe rear impact followed by a less severe but still significant frontal impact – matched crash scenario.
- Conclude data **is** from the event of interest based on Delta V matching up with facts.

How much data do I have?

- The EDR had 6 data points of speed, brake, accel pedal and RPM – 5@1 sec intervals and the 6th data point at first event TRG.
- Longitudinal Delta V – Single Peak Value 53.1 mph (85.5 kph) for rear impact and graph and data table for 200ms at 10ms intervals
- Second set of Delta V data for frontal impact that followed rear impact.

What Can I do with it?

- $S_C = \Delta V_1 * \frac{W1+W2}{W2}[\frac{1}{1+e}]$
  Use Rear Impact Delta V to get closing speed
- $S_C = 53.1 * \frac{2286+5687}{5687}[\frac{1}{1+5\%}]$

- Closing Speed S_C = 70.7 mph (113.8 kph)
- Range +/-10%, 63.6 to 77.8 mph

Porsche Speed at First Impact

- Impact Speed= Closing Spd + Toyota Initial Spd
- V_1 = 70.7 mph closing speed + 1.2mph Yaris
- V_1 = 71.9 mph (115.8 kph)

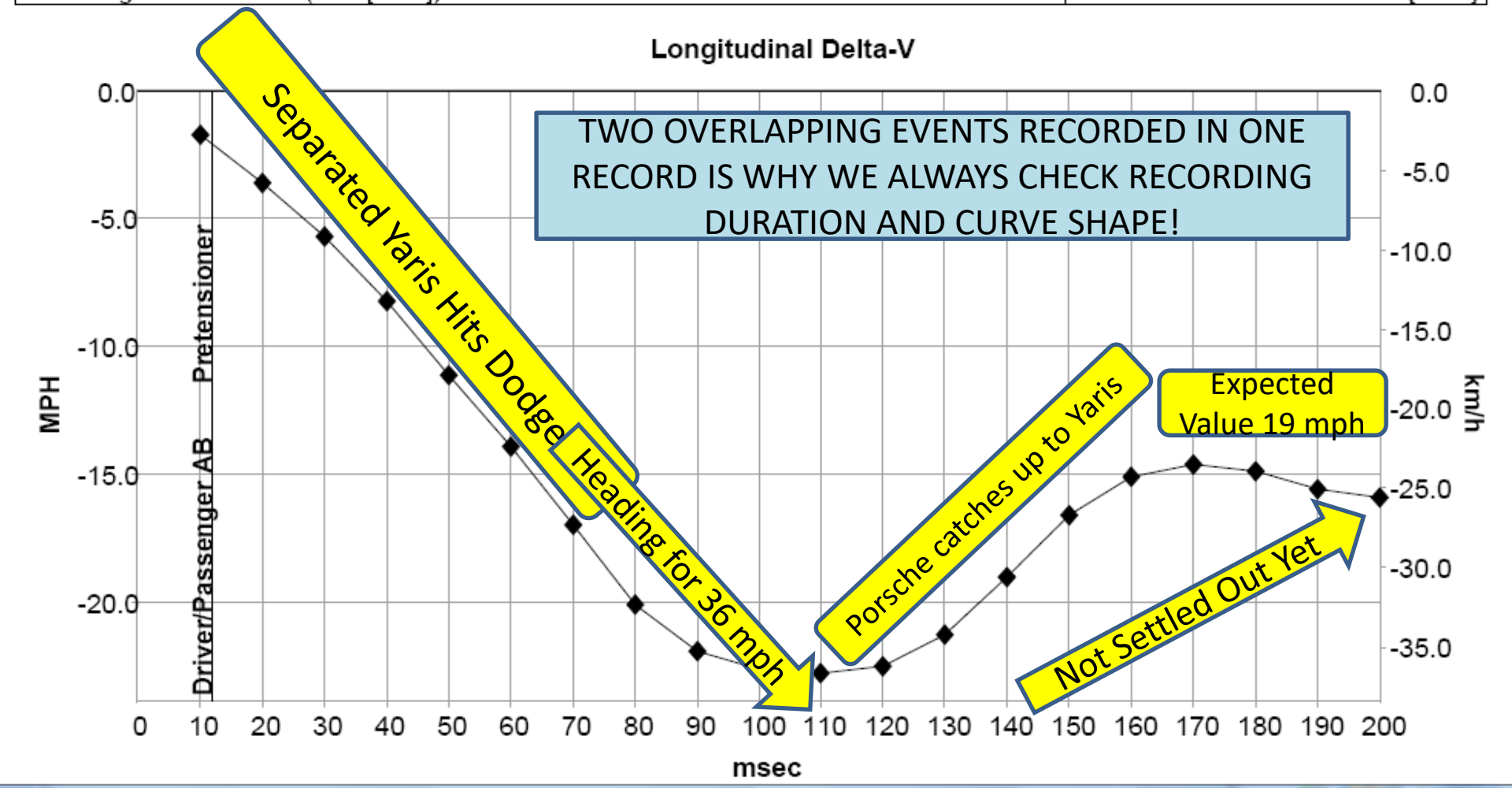
Speed at Second Impact

- Porsche and Toyota depart first impact at 54.3 mph. They may separate slightly but will recouple as soon as Toyota gets into back of Dodge. Treat second collision as combined mass of Porsche & Toyota hitting Dodge.
- (5687+2286)*54.3mph = (5687+2286+4472) * Departure Speed
- Common Departure Speed = 34.7 mph
  (based on EDR 1st impact DV alone)
- Second Impact Toyota Delta V, based on the above, would be expected to be 54.3 impact -34.7 departure = 19.6 mph Delta V.

Second Impact EDR Delta V Trace

Longitudinal Crash Pulse (Most Recent, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH[km/h])	-22.8 [-36.7]



Reconcile EDR with Scene Evidence

- Porsche Speed from Toyota ΔV = 71.7 mph
- Scene Evidence Slide to Stop and Momentum Calculation = 79 mph Porsche speed at impact
- Good agreement – Delta V is +/-10% & often understated. **CLIPPING is a likely explanation** for why the EDR is low in this case.

Toyota EDR Accuracy

- Toyota stated up front that its first prototype readout tool was not validated –but that didn't mean it wasn't right.
- Because early Toyota EDR's (like many other manufacturers) that lost power at impact may not make a new recording at impact, there were examples of when the Toyota EDR's did not match the accident facts.
- Some claimed the Toyota EDR was unreliable - based on a lack of understanding of the not-yet-written data limitations.

Specific Toyota Validation

- Niehoff 2005: 3 barrier crashes: Delta V ranged -6.7% to +1.1%, only vehicle with speed data was -0.9% of reference instrument.
- Gabler 2008: 11 frontal/4 side barrier crashes: Delta V data ranged from -22% to +11%, avg -1.6%. Only one vehicle had speed data, it reported 35 mph in a 33.8 mph test or 3% high. Belt status was reported correctly in all 15 tests
- NHTSA 2011: 14 staged events, “No errors in measurement and storage or pre crash data” (within known accuracy and timing limitations). Vehicle speed within 1.5 mph, throttle voltage within 2.5%.

Additional Toyota Validation

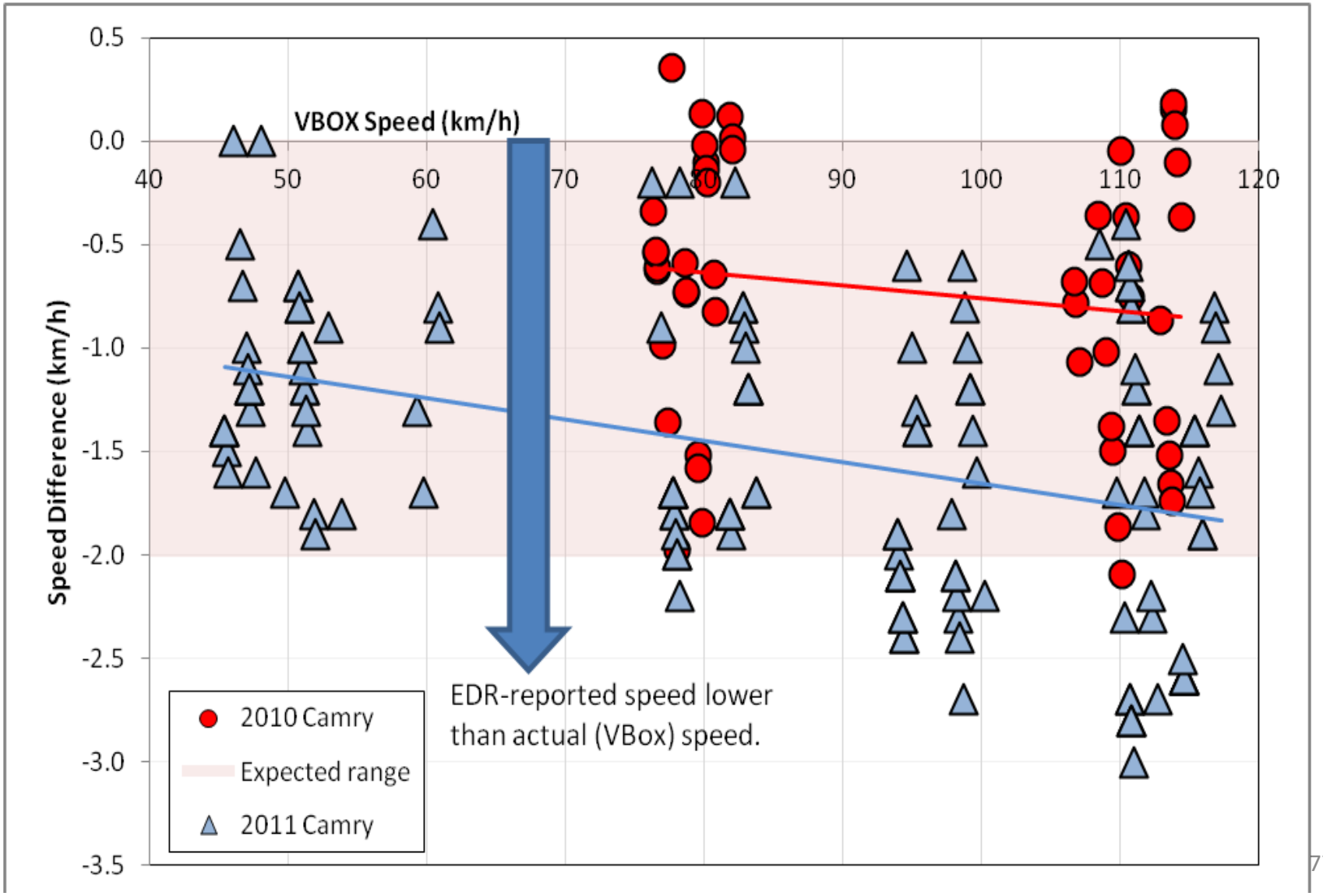
- Ishikawa & Takubo 2009: 14 J-NCAP crashes EDR Delta V is from -14% to +0.5% of reference instrumentation, averaging -7.3%. Speed data within 4%, avg. 2%
- Ruth 2012: SAE 2012-01-0999 -Speed data within 3%
- Brown 2012: SAE 2012-01-0996 & 0998 (Actual speed -2.7%)
- Wilkinson 2013: SAE 2013-01-1268 (Gen 1 Delta V 1mph low in low speed frontal collisions, 1mph high in rear collisions)
- Webster 2014: SAE 2014-01-0505 **Accuracy of Recorded Driver Inputs in Toyota Part 563 EDR** (Only addressed accuracy of brake and accelerator pedal position).
- A new paper in peer review for 2016 suggests Gen 2 ACM Delta V may be under reported slightly in frontal collisions, and over reported slightly in rear collisions, similar Wilkinson's results on Gen 1's above.

CONCLUSION

When the recording is from the event of interest, and data limitations are understood, there is no reason to believe speed data is any less accurate than any other manufacturer's EDR based on 42 test events. Delta V may under report slightly in Gen 1 and Gen 2 ACM's in frontal collisions and under reported in rear collisions, but in barrier tests are still within the generally accepted +/-10%. Other than the 00 EDR seat belt reporting at time of readout and not event noted in the data limitations, there are no known issues with other data elements at this time.

Toyota Speed Data Accuracy Validated

SAE 2012-01-0999 2010/11 Camry



The Accuracy and Sensitivity of 2005 to 2008 Toyota Corolla Event Data Recorders in Low-Speed Collisions
SAE 2013-01-1268

- Induced a series of low speed events on vehicles and on a sled
- Verified 2G wakeup threshold thru testing
- Determined “offset” in Toyota accelerometers causing them to read 1.3 to 2.6 km/h low in frontal collisions and 0.6 to 2.2 km/h high in rear collisions.

Gen 1 vs Gen 2 EDR Data

Data	00/02 Gen 1	04/06 Gen 2
# of Events	3 frontal events, 1 side (if equipped), 1 roll (if equipped)	2 front, 2 side (if equipped), 2 roll (if equipped)
Recording Priority	Lock deployments, If no deploy, 2 most recent 3rd = most negative	Lock Deployments If no deploy, 2 most recent
Longitudinal Delta V 0 = TRG	15 @ 10ms = 150ms 0 = TRG	20 @ 10ms = 200ms 0 = TRG
Polarity:	Positive for Front Impact	Negative for Front Impact
Lateral Delta V (if equipped) 0 = TRG	14 @ 6ms = 84 ms -24 to +54 ms	25 @ 4ms = 100ms -20 to +76 ms
Polarity:	?	?
Roll Angle & Lateral G (if equipped)	20 @ ?? ms interval	20 @ 125 ms interval -625ms to +1800ms
PRE CRASH		
Speed, Brake, Throttle, RPM	6, 5 @ 1ms, last @ TRG	6, 5 @ 1ms, last @ TRG
Resolution: Speed	2km/h (1.2 mph)	2km/h (1.2 mph)
Resolution: Brake	On/Off	On/Off
Resolution: Throttle	Off/Middle/Full Volts 0.78 to 3.71	Off/Middle/Full Volts 0.78 to 3.71
Resolution: RPM	400	400

Gen 1 Mini Inline Case Review

- **Longitudinal Delta V** was -**22.2 mph (-35.7 kph)**
- After impact the vehicle slid 15 feet with 25% braking on a 0.7 drag factor surface, yielding a **post impact speed** of **9mph (15 kph)**.
- The departure angle was 45 degrees.
- Based on the EDR Delta V and these facts, ***what was the impact speed***?

Mini Inline Case Solution

- **Departure Speed in X direction**
- V_{3x} = 9 mph * cosine 45 degrees
- V_{3x} = 9mph * .707 = **6.3 mph (10.1 kph)**
- Approach = 6.3 – (Delta V_x)
- Approach = 6.3 - (-22.2)
- Approach = **28.5 mph (45.9 kph)**
- **+/-10% of ΔV** = +/-2.9 **= 25.6 to 31.4 mph**
- Reference last EDR **precrash** speed was **27.3mph**

2013 Toyota “12EDR”RAV4 Project



Crash Scenario

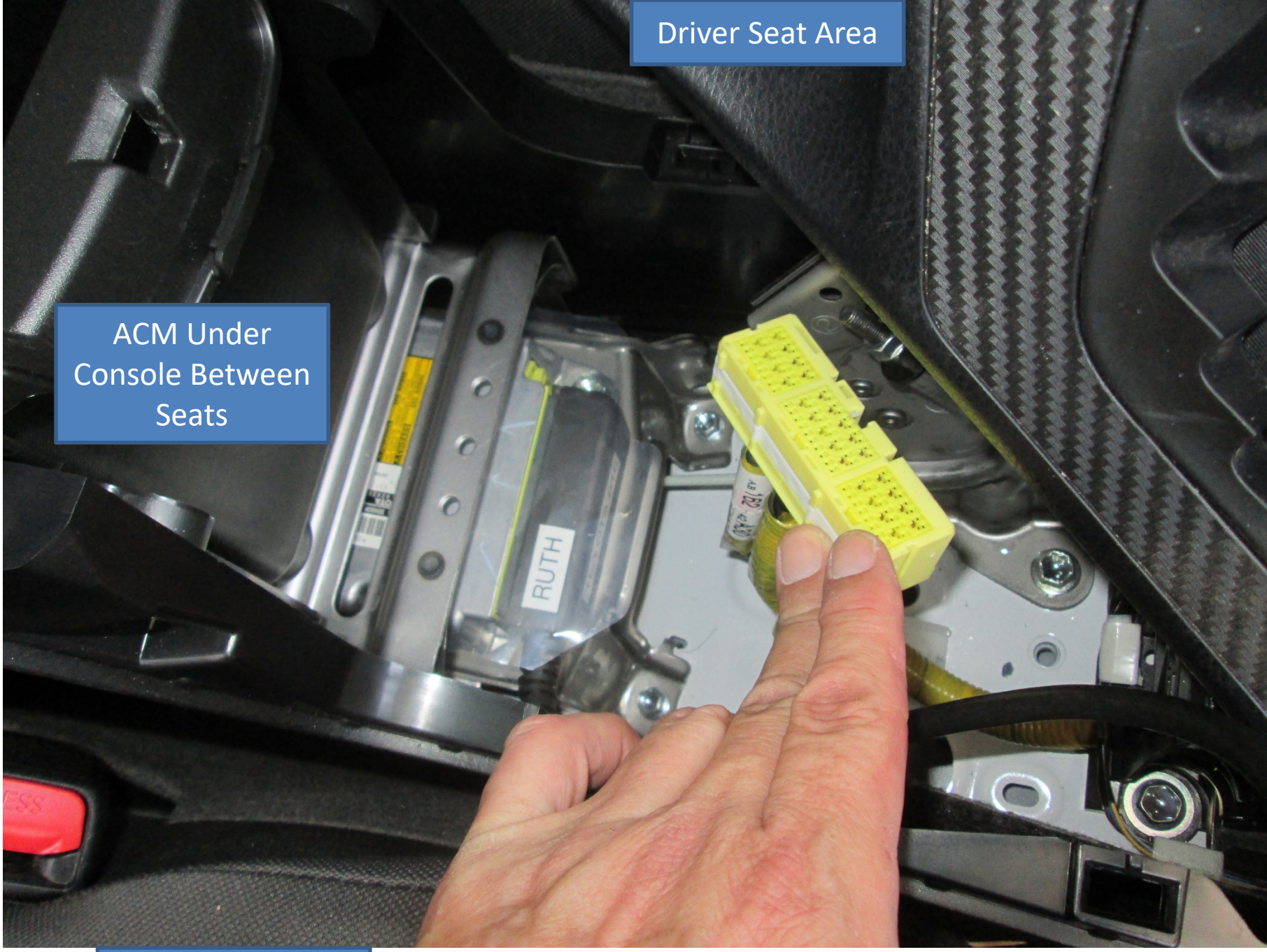
- The Rav4 and an 2012 BMW 3 series are coming in opposite directions on a rural 2 lane 50 mph road at night. They collide at an intersection on the Rav4's fog line. The Rav 4 driver got a concussion and can't remember the crash but should have been going straight thru the intersection. His passenger was sleeping and did not see the crash.
- Rav4's family is suing Mr. BMW. The BMW driver now claims the Rav4 crossed centerline into his lane, and that to avoid the Rav4 he crossed centerline to the open lane, but at the last minute the Rav4 swerved back into his own lane.
- The police investigation was not very thorough. No pictures were taken. V2's family suspects V1 was drunk but no charges were filed, V1 was simply ticketed for being out of his lane.
- There are very few facts to go on, V2's EDR may be the only objective witness. V2's attorney has asked you to examine it and determine what you can learn about the crash from it.

Learning Objectives

- Learn Part 563 intent Toyota “12 EDR” Data
- Determine which event(s) in a multi-event report are relevant to our crash
- Determine PDOF from lateral and longitudinal ΔV
- For the precrash data just prior to impact, determine if the Rav4 swerved, if so how many degrees it swerved, and how many feet sideways it moved from its original lane (did it cross centerline).
- From the Rav4 Delta V data and scene evidence, determine speed at impact.
- From the Rav4 Delta V data and vehicle weights, determine closing speed and approximate speed of the other vehicle.
- From all the data available, working time distance, determine who you believe was at fault in the crash

Airbags & Dr. Kneebag Deployed




Driver Seat Area
ACM Under Console Between Seats
Pass Seat Area

2013 Toyota RAV4 Multi Event Crash

CDR File Information

User Entered VIN	JTMBFREV4D5******
User	TDB
Case Number	RAV4
EDR Data Imaging Date	05/08/2014
Crash Date	05/07/2014
Filename	2013 RAV4 PROJECT FILE.CDRX
Saved on	Thursday, May 8 2014 at 15:04:43
Collected with CDR version	Crash Data Retrieval Tool 12.3
Reported with CDR version	Crash Data Retrieval Tool 12.3
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (1), Side (2), Rollover (1)

4 Events!!

2013 Toyota RAV4

System Status at Time of Retrieval

ECU Part Number		89170-42420
EDR Generation		12EDR
Complete File Recorded	Caution!!!	No
Freeze Signal		ON
Freeze Signal Factor		Front Airbag Deployment
Diagnostic Trouble Codes Exist		No
Ignition Cycle ,Download (times)		1409
Multi-event, number of events (times)	Beware incomplete recording	N/A
Time from event 1 to 2 (s)		N/A
Time from Previous Pre Crash	Beware incomplete recording	16381 or greater
Latest Pre-Crash Page		0
Contains Unlinked Pre-Crash Data		No

Examine event numbers and completion status – only rollover is incomplete – does that mean it was last? If so where did #3 go?? Check each event for prior event type!!

#1, #2, #4, and N/A may be from the same crash!!! Caution!!!

Lost due to incomplete rec

One set only

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	4	Side Crash	0	Complete (Page 0)	Complete (Side Page 1)
1st Prior Event	2	Front/Rear Crash	N/A	N/A	Complete (Front/Rear Page 0)
2nd Prior Event	1	Side Crash	N/A	N/A	Complete (Side Page 0)
TRG Invalid #1	N/A	Rollover	N/A	N/A	Incomplete (Rollover Page 0)

So the Rollover is #3 – it STARTED before #4 but because rollovers record POST impact, it didn't FINISH before #4!

System Status at Event (Most Recent Event, TRG 4)

Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	4
Previous Crash Type	Rollover
Time from Pre-Crash TRG (msec)	96
Linked Pre-Crash Page	0
Side Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Side Curtain Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Pretensioner Deployment, Time to Fire (msec)	No
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

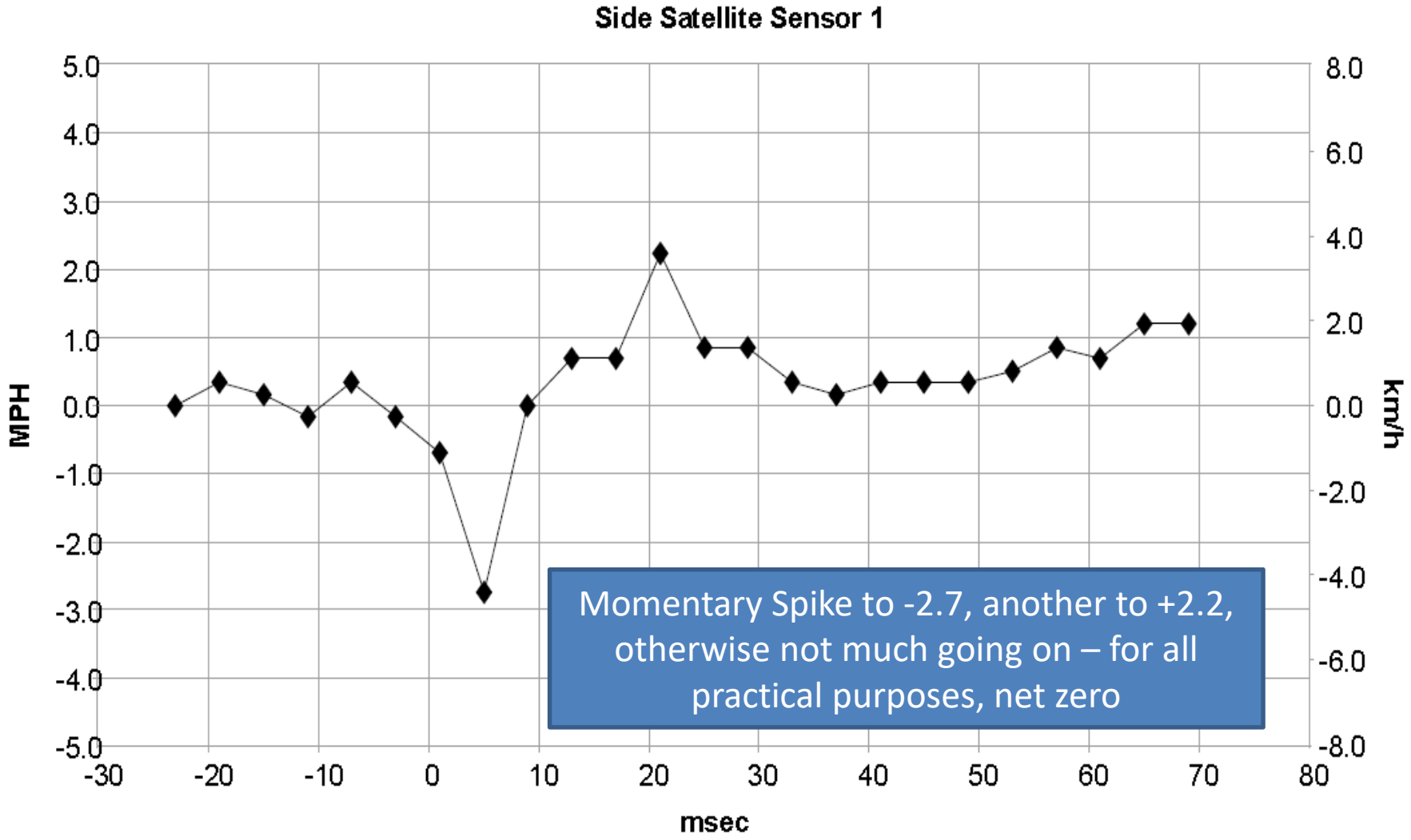
Roll Start 0.1 sec before last side event

The last side event was small

Lateral Crash Pulse (Most Recent Event, TRG 4 - table 1 of 2)

Recording Status, Time Series Data	Complete
Recorded Side	Left Side
Time from TRG to Next Sample (msec)	1
Location of Side Satellite Sensor 1	Front Door
Location of Side Satellite Sensor 2	Not Equipped
Location of Side Satellite Sensor 3	Not Equipped
Location of Side Satellite Sensor 4	C-Pillar
Maximum Delta-V Lateral, Side Satellite Sensor 1 (MPH [km/h])	-2.7 [-4.4]
Maximum Delta-V Lateral, Side Satellite Sensor 2 (MPH [km/h])	N/A
Maximum Delta-V Lateral, Side Satellite Sensor 3 (MPH [km/h])	N/A
Maximum Delta-V Lateral, Side Satellite Sensor 4 (MPH [km/h])	4.6 [7.4]

Is this from after the initial crash (after rollover start?)



DTCs Present at Time of Event (Most Recent Event, TRG 4)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 4)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	500
Safety Belt Status, Driver	ON
Safety Belt Status, Front Passenger	ON
Occupant Size Classification, Front Passenger	AM50 (Not Child)
Frontal Airbag Suppression Switch Status, Front Passenger	SNA
RSCA Disable Switch	SNA
Seat Track Position Switch, Foremost, Status, Driver	No
Airbag Warning Lamp, On/Off	OFF
Ignition Cycle ,Crash (times)	1403

Precrash Data For Side Crash #4

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 4)

Time (sec)	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	50.3 [81]	50.3 [81]	48.5 [78]	45.4 [73]	41.6 [67]	36.7 [59]	29.8 [48]	22.4 [36]	13 [21]	7.5 [12]	8.1 [13]
Accelerator Pedal, % Full (%)	18.5	8.0	0.0	0.0	0.0	0.0	0.0	4.5	17.0	0.0	0.0
Percentage of Engine Throttle (%)	5.5	5.5	0.0	0.0	0.0	0.0	0.0	0.0	5.0	3.5	0.0
Engine RPM (RPM)	1,500	1,500	1,400	1,300	1,200	1,100	900	800	800	1,300	1,400
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	OFF	OFF
Brake Oil Pressure (Mpa)	0.00	0.00	2.11	2.35	3.36	5.81	5.86	7.87	9.31	0.00	0.00
Longitudinal Acceleration , VSC	-0.215	-0.287	-3.230	-3.015	-4.020	-6.173	-6.604	-7.968	-8.973	-0.287	-3.589
Yaw Rate (deg/sec)	-1.46	-1.46	-0.49	-0.98	-0.98	-1.46	-0.98	-0.49	-1.95	-11.71	-17.57
Steering Input (degrees)	-3	-3	-3	-3	-3	-3	-3	-3	-30	-174	-204
Shift Position											
Sequential Shift Range											
Cruise Control Status											
Drive Mode, PWR											
Drive Mode, ECO											
Drive Mode, Sport											

1. Braking pressure and Long Accel rate build -4 to -1, go over 50% AT -2.5
3. Speed loss -1.5 to -1.0 9.6 mph over 0.5 sec, 0.9G equiv matches long accel
4. Steering + yaw shows slight right input before hard right swerve in last 1 second

Yaw and Steering are opposite SAE polarity

Data Element Sign Convention:

The following table provides an explanation of the sign notation for data elements that may be inclu

Data Element Name	Positive Sign Notation Indicates
Max. Longitudinal Delta-V	Forward
Longitudinal Delta-V	Forward
Maximum Delta-V Lateral, Side Satellite Sensor 1	Left to Right
Maximum Delta-V Lateral, Side Satellite Sensor 2	Left to Right
Maximum Delta-V Lateral, Side Satellite Sensor 3	Left to Right
Maximum Delta-V Lateral, Side Satellite Sensor 4	Left to Right
Delta-V Lateral, Side Satellite Sensor 1	Left to Right
Delta-V Lateral, Side Satellite Sensor 2	Left to Right
Delta-V Lateral, Side Satellite Sensor 3	Left to Right
Delta-V Lateral, Side Satellite Sensor 4	Left to Right
Roll Angle Peak	Clockwise Rotation
Roll Angle at the Time of TRG	Clockwise Rotation
Roll Rate	Clockwise Rotation
Lateral Acceleration , Airbag ECU Sensor *	Left to Right
Longitudinal Acceleration , VSC Sensor	Forward
Yaw Rate	Left Turn
Steering Input	Left Turn

* For sensing a rollover

Trigger 2: The “Big” Frontal?

System Status at Event (1st Prior Event, TRG 2)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	2
Previous Crash Type	Side Crash
Time from Pre-Crash TRG (msec)	2
Linked Pre-Crash Page	0
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (msec)	2
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front Passenger (msec)	2
Pretensioner Deployment, Time to Fire, Driver (msec)	2
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	2
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	32
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (msec)	12
Active Head Restraint, Time to Deploy, Driver (msec)	SNA
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	2
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec)	2
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

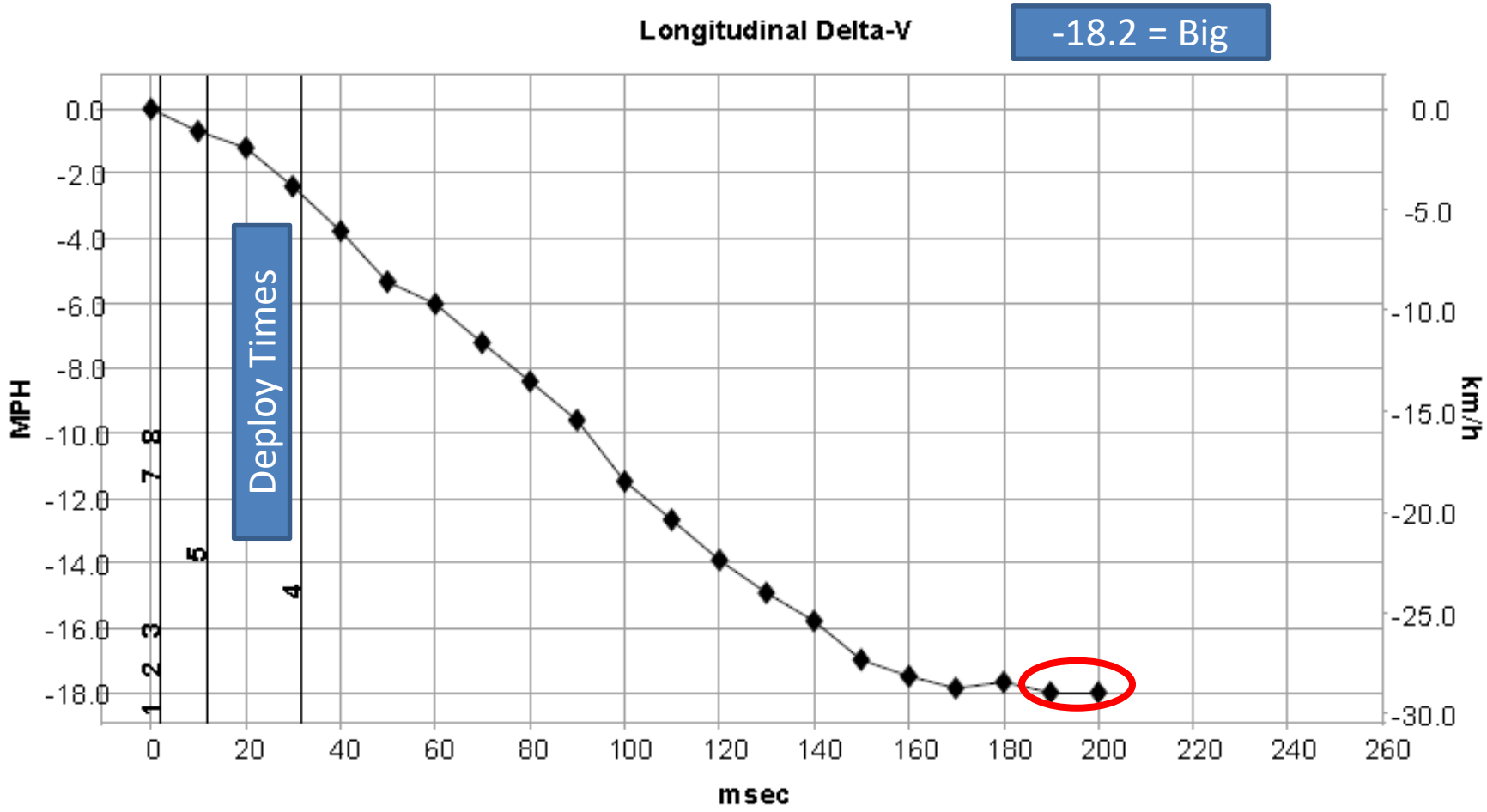
Said N/A IN SUMMARY

FAST!!

Stage 2 = Big

Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	2.0
Length of Delta-V (msec)	200
Max. Longitudinal Delta-V (MPH [km/h])	-18.2 [-29.2]
Time, Maximum Delta-V, Longitudinal (msec)	201.0
Power Supply Status at Max. Delta-V	ON



Data Table for Prior Graph

Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Power Supply Status
0	0.0 [0.0]	ON
10	-0.7 [-1.1]	ON
20	-1.2 [-1.9]	ON
30	-2.4 [-3.9]	ON
40	-3.8 [-6.1]	ON
50	-5.3 [-8.6]	OFF
60	-6.0 [-9.7]	OFF
70	-7.2 [-11.6]	OFF
80	-8.4 [-13.5]	OFF
90	-9.6 [-15.4]	OFF
100	-11.5 [-18.5]	OFF
110	-12.7 [-20.4]	OFF
120	-13.9 [-22.3]	ON
130	-14.9 [-24.0]	ON
140	-15.8 [-25.4]	ON
150	-17.0 [-27.3]	ON
160	-17.5 [-28.1]	ON
170	-17.8 [-28.7]	ON
180	-17.7 [-28.4]	ON
190	-18.0 [-29.0]	ON
200	-18.0 [-29.0]	ON
210	0.0 [0.0]	ON
220	0.0 [0.0]	ON
230	0.0 [0.0]	ON
240	0.0 [0.0]	ON
250	0.0 [0.0]	ON

Note NOT -18.2

Algorithm Back to Sleep – grayed out

System Status at Event (2nd Prior Event, TRG 1)

Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Side Airbag Deployment, Time to Deploy (If Equipped) (msec)	No
Side Curtain Airbag Deployment, Time to Deploy (If Equipped) (msec	No
Pretensioner Deployment, Time to Fire (msec)	No
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA

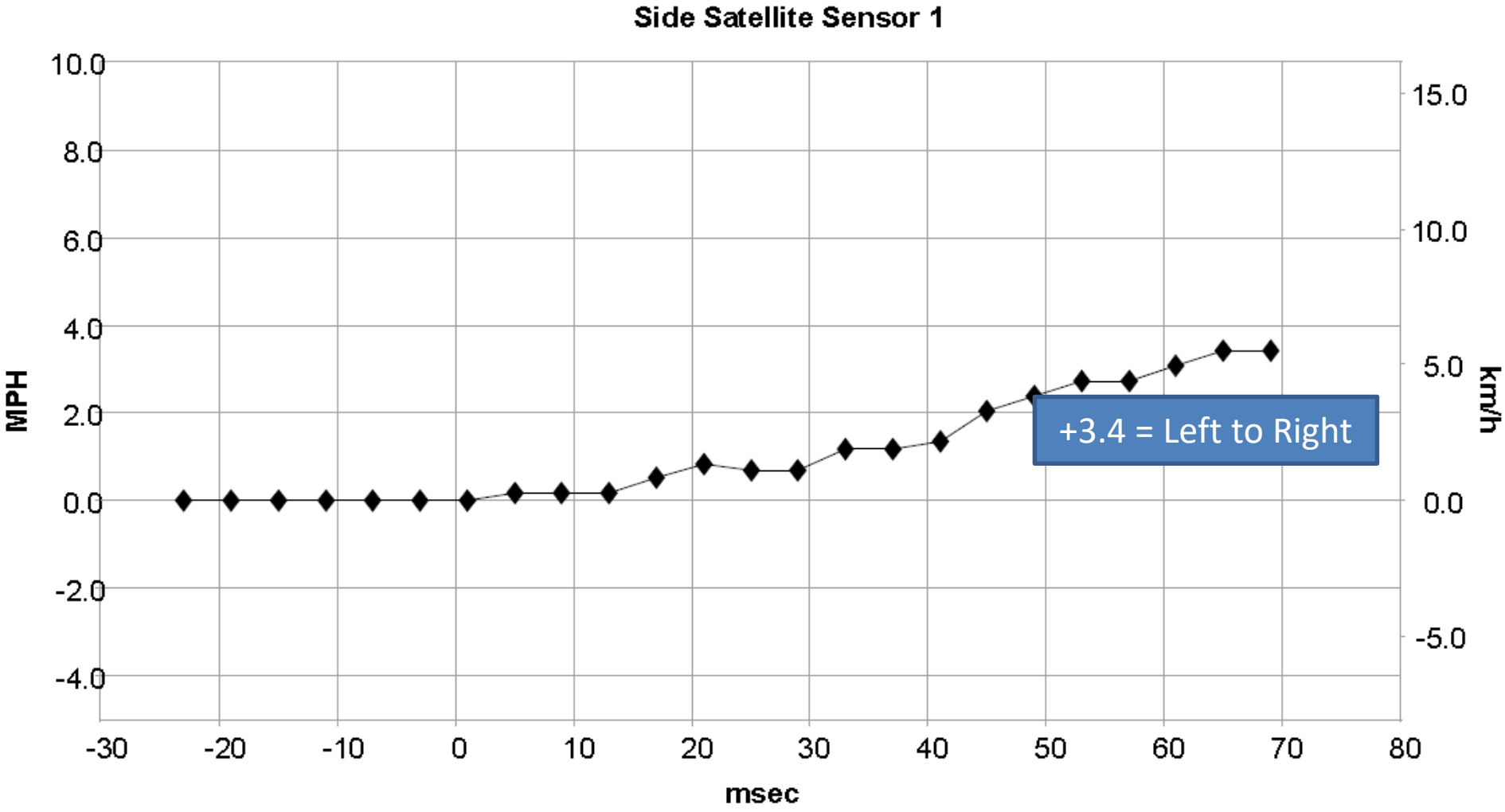
Said N/A IN SUMMARY

So was this related or not????

Lateral Crash Pulse (2nd Prior Event, TRG 1 - table 1 of 2)

Recording Status, Time Series Data	Complete
Recorded Side	Left Side
Time from TRG to Next Sample (msec)	1
Location of Side Satellite Sensor 1	Front Door
Location of Side Satellite Sensor 2	Not Equipped
Location of Side Satellite Sensor 3	Not Equipped
Location of Side Satellite Sensor 4	C-Pillar
Maximum Delta-V Lateral, Side Satellite Sensor 1 (MPH [km/h])	3.4 [5.5]
Maximum Delta-V Lateral, Side Satellite Sensor 2 (MPH [km/h])	N/A
Maximum Delta-V Lateral, Side Satellite Sensor 3 (MPH [km/h])	N/A
Maximum Delta-V Lateral, Side Satellite Sensor 4 (MPH [km/h])	2.7 [4.3]

Side event 1 part of Frontal Event 2?



Data Table for prior graph

Lateral Crash Pulse (2nd Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Delta-V Lateral, Side Satellite Sensor 1 (MPH [km/h])	Delta-V Lateral, Side Satellite Sensor 2 (MPH [km/h])	Delta-V Lateral, Side Satellite Sensor 3 (MPH [km/h])	Delta-V Lateral, Side Satellite Sensor 4 (MPH [km/h])
-23	0.0 [0.0]	SNA	SNA	0.0 [0.0]
-19	0.0 [0.0]	SNA	SNA	0.0 [0.0]
-15	0.0 [0.0]	SNA	SNA	0.0 [0.0]
-11	0.0 [0.0]	SNA	SNA	0.0 [0.0]
-7	0.0 [0.0]	SNA	SNA	0.0 [0.0]
-3	0.0 [0.0]	SNA	SNA	0.0 [0.0]
1	0.0 [0.0]	SNA	SNA	0.0 [0.0]
5	0.2 [0.3]	SNA	SNA	0.0 [0.0]
9	0.2 [0.3]	SNA	SNA	0.1 [0.1]
13	0.2 [0.3]	SNA	SNA	0.3 [0.4]
17	0.5 [0.8]	SNA	SNA	0.3 [0.6]
21	0.9 [1.4]	SNA	SNA	0.3 [0.6]
25	0.7 [1.1]	SNA	SNA	0.3 [0.6]
29	0.7 [1.1]	SNA	SNA	0.6 [1.0]
33	1.2 [1.9]	SNA	SNA	0.7 [1.1]
37	1.2 [1.9]	SNA	SNA	1.0 [1.7]
41	1.4 [2.2]	SNA	SNA	1.2 [1.9]
45	2.1 [3.3]	SNA	SNA	1.4 [2.2]
49	2.4 [3.9]	SNA	SNA	1.6 [2.6]
53	2.7 [4.4]	SNA	SNA	1.9 [3.0]
57	2.7 [4.4]	SNA	SNA	2.2 [3.6]
61	3.4 [5.5]	SNA	SNA	2.5 [4.0]
65	3.4 [5.5]	SNA	SNA	2.6 [4.1]
69	3.4 [5.5]	SNA	SNA	2.7 [4.3]

Rollover Listed as “Invalid trigger”

System Status at Event (TRG Invalid #1)

Recording Status, Rollover Crash Info.	Incomplete
Crash Type	Rollover
TRG Count (times)	Invalid
Previous Crash Type	Front/Rear Crash
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Side Curtain Airbag Deployment, Time to Deploy (msec)	0
Pretensioner Deployment, Time to Fire, Driver (msec)	0
Pretensioner Deployment, Time to Fire, Front Passenger (msec)	0

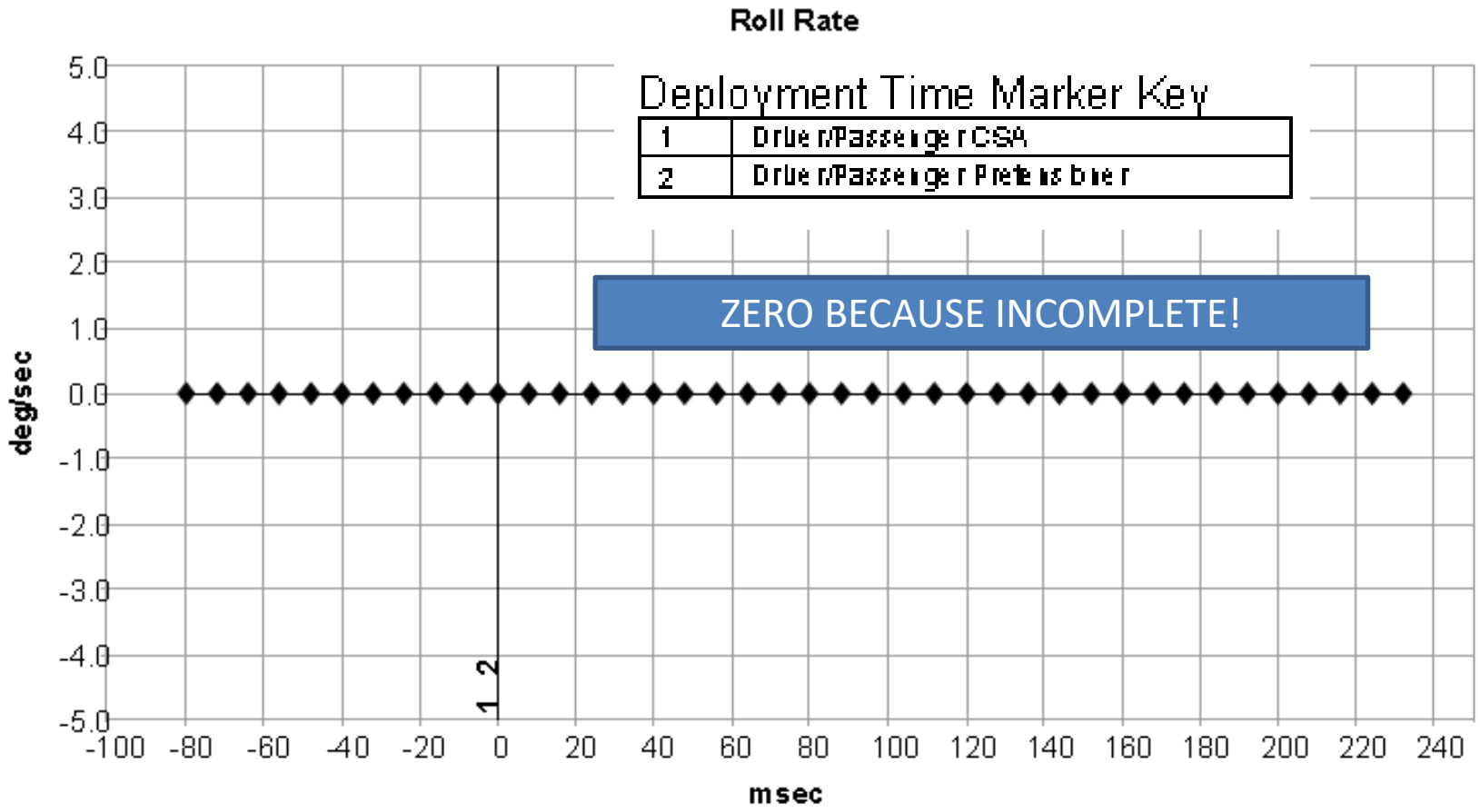
EVENT 3

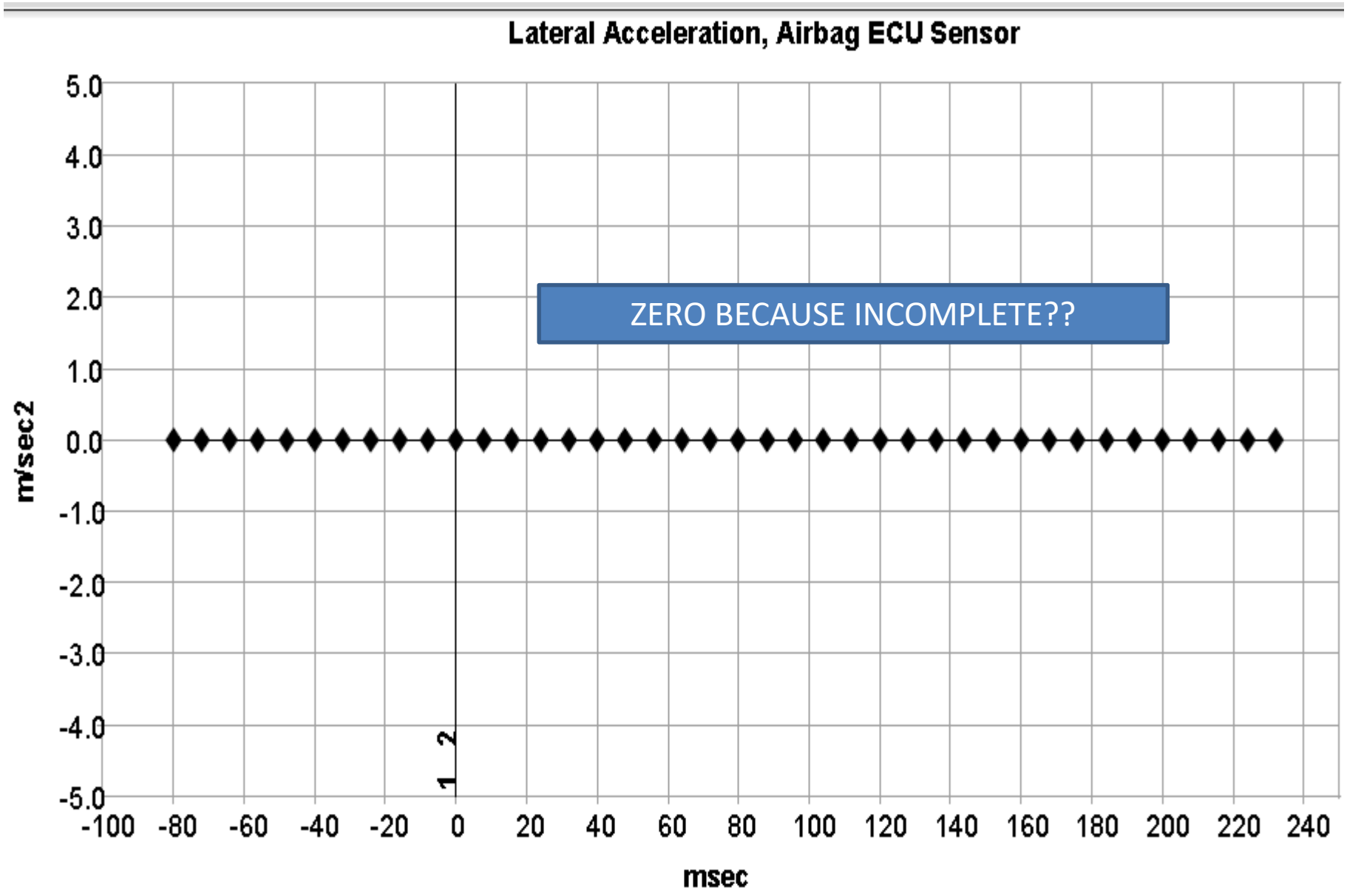
EVENT 2

When did it roll over????

Rollover Crash Pulse (TRG Invalid #1 - table 1 of 2)

Recording Status, Time Series Data	Incomplete
Time from TRG to Next Sample (msec)	0
Roll Angle Peak (degrees)	0.0
Roll Angle at the Time of TRG (degrees)	0.0



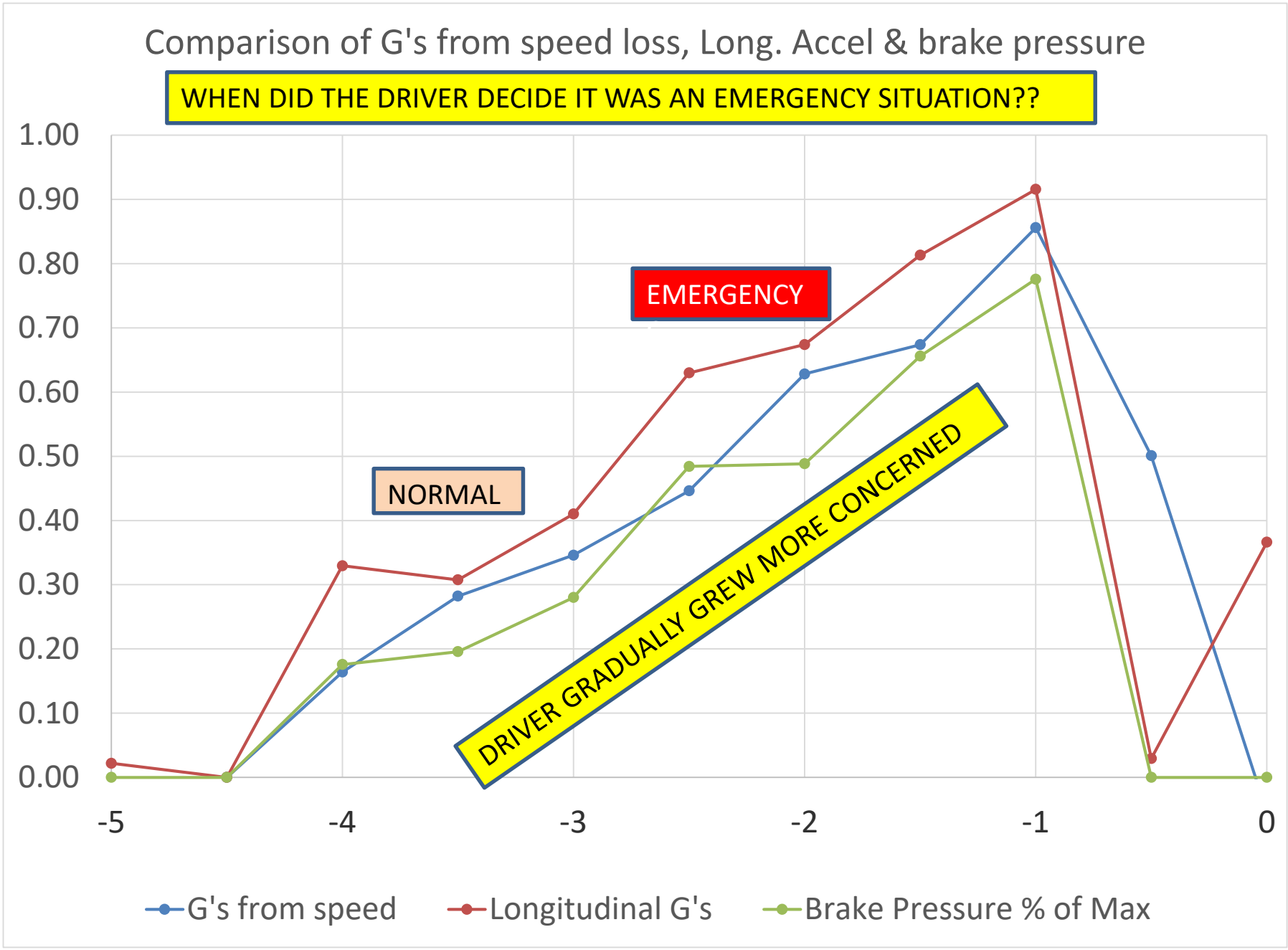

Lateral Acceleration, Airbag ECU Sensor
ZERO BECAUSE INCOMPLETE??
m/sec2
msec
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-100
-80
-60
-40
-20
0
20
40
60
80
100
120
140
160
180
200
220
240
1_2

Learnings so far:

- All 4 events are related and are linked to precrash data Page “0” and so must be within 0.5 seconds of each other. While the summary does not say they are linked to Page 0, the individual pages do.
- The rollover is not numbered but must Trigger 3, it starts after the main crash on the driver front corner. TRG4 is set after the rollover begins, but because it has very little net Delta V, we may never know what physical event in the postcrash movement set this trigger.
- The precrash steering and yaw data do not show any left input in the 5 seconds before the crash, but do show a hard right swerve prior to impact that would also fit the BMW driver’s story.
- The RAV4 is slowing at an increasing rate, but why if they are going straight thru a clear intersection??

Data Analysis

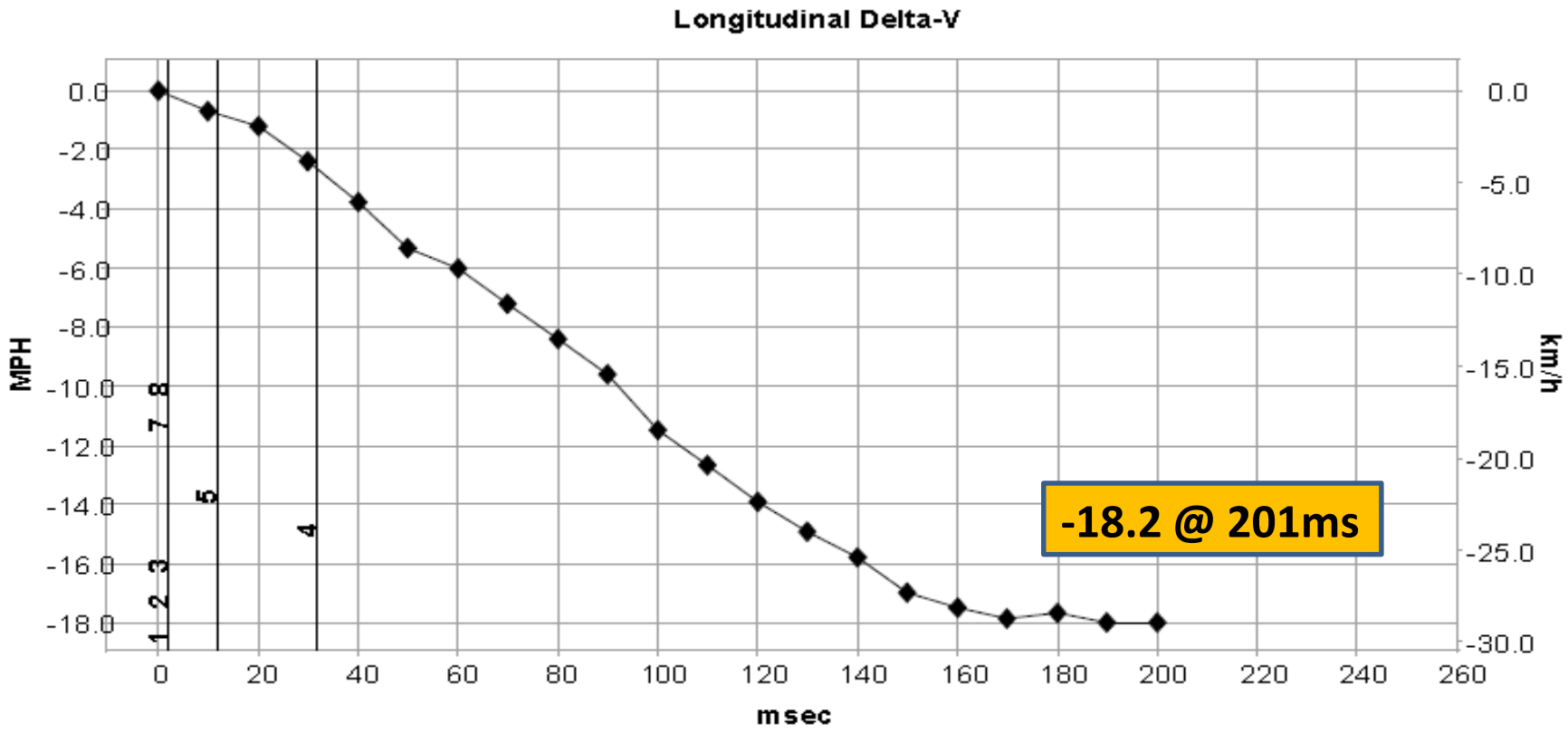
- Take consecutive speed data points, subtract them to get speed change vs time, divide by accel due to gravity to get how many G's. For -1.5 sec:
- (29.8-22.4) = 7.4 mph*1.466/32.2/0.5 sec = 0.67G
- Compare this to longitudinal acceleration in m/s, divide by accel due to gravity of 9.8 m/sec2 to get G's -7.968m/sec2 / 9.8 = 0.82G
- Compare this to % max brake pressure in Mpa 7.87/12 = 66%
- See next slide for graph


Comparison of G's from speed loss, Long. Accel & brake pressure
WHEN DID THE DRIVER DECIDE IT WAS AN EMERGENCY SITUATION??
1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
-5
-4
-3
-2
-1
0
EMERGENCY
NORMAL
DRIVER GRADUALLY GREW MORE CONCERNED
G's from speed
Longitudinal G's
Brake Pressure % of Max

Delta V analysis – What's the PDOF? The Total??

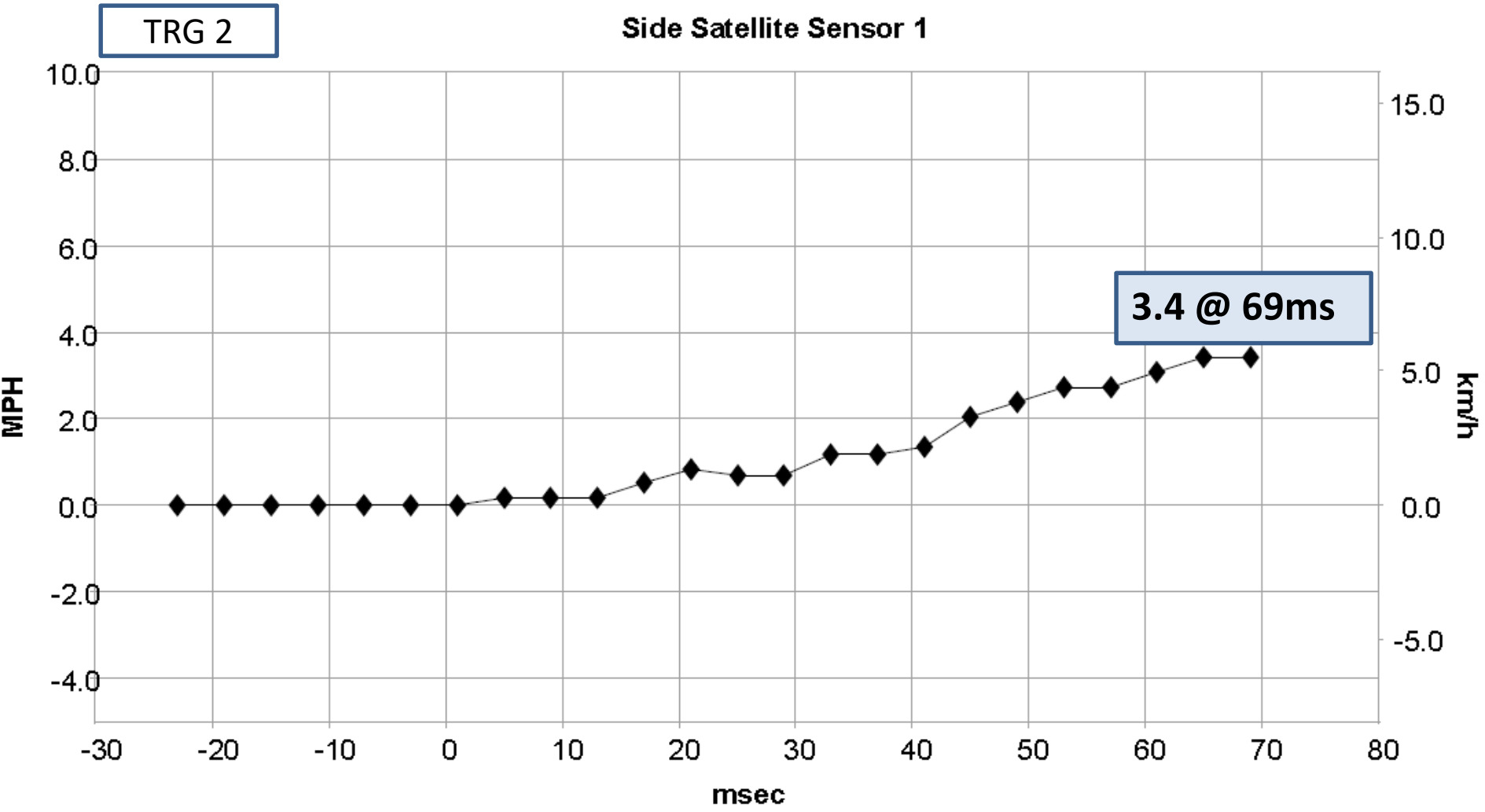
Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)

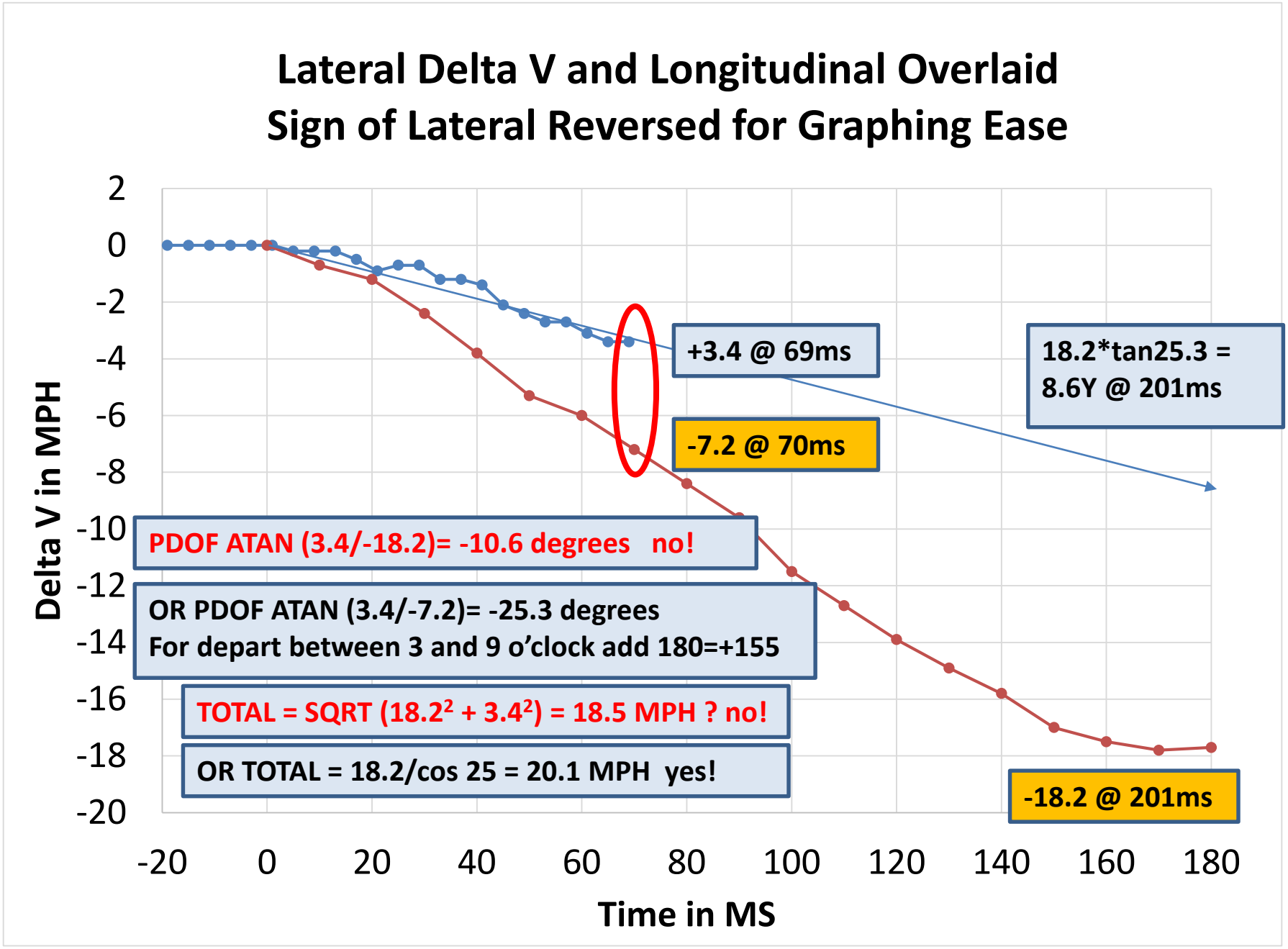
Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	2.0
Length of Delta-V (msec)	200
Max. Longitudinal Delta-V (MPH [km/h])	-18.2 [-29.2]
Time, Maximum Delta-V, Longitudinal (msec)	201.0
Power Supply Status at Max. Delta-V	ON



Lateral Delta V

Gen 3 does NOT report lateral DV from the central control module – only from satellites. Use closest to CG




Lateral Delta V and Longitudinal Overlaid
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees
For depart between 3 and 9 o'clock add 180=+155
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

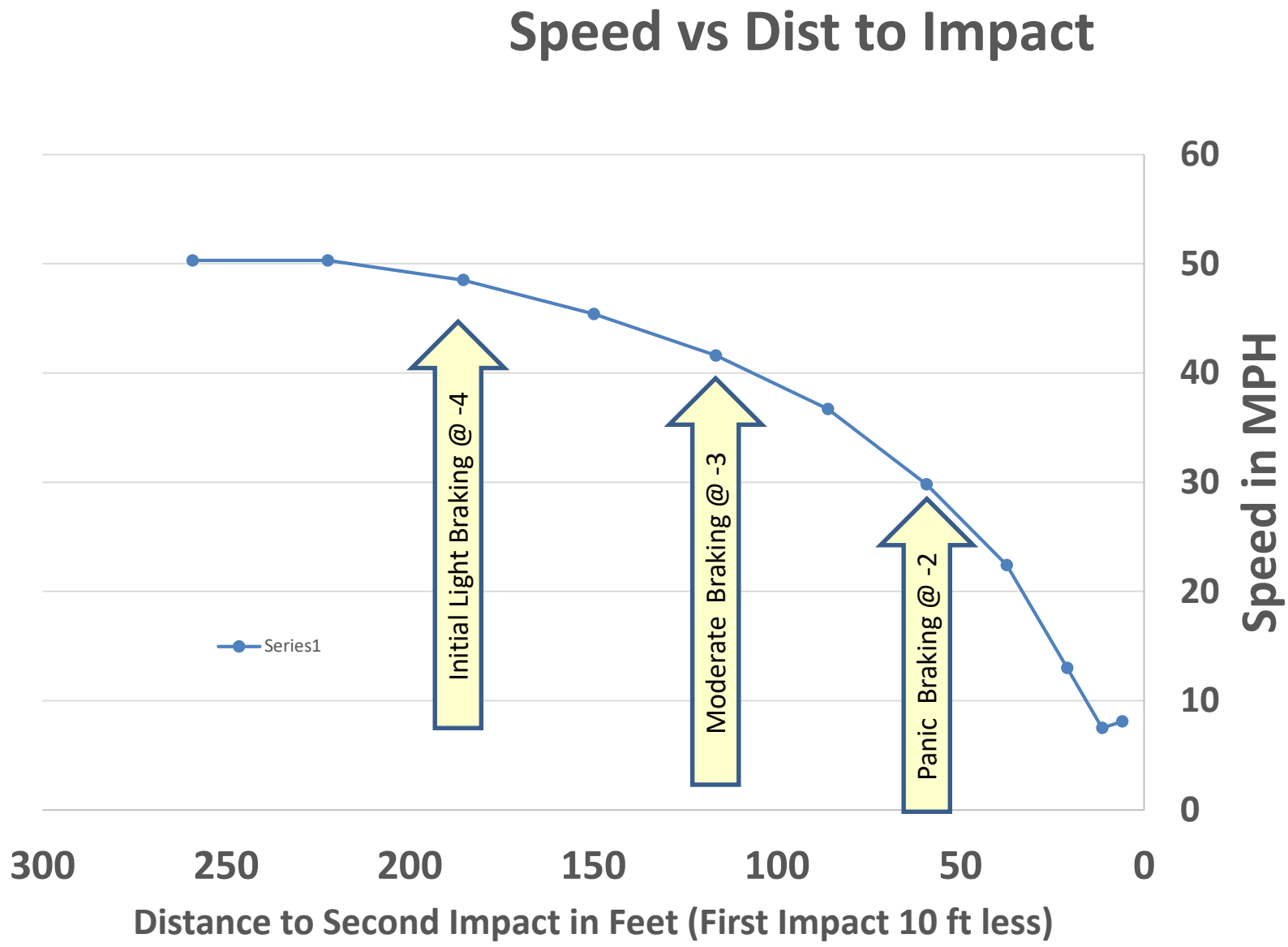
How much does RAV4 Swerve?

- Swerve starts at -1
- Avg rate -1 to -0.5, (1.96+11.71)/2 = 6.8^{o}/sec
- Heading change .5 sec * 6.8^{o}/sec = 3.4 deg
- Avg rate -0.5 to 0.0 = (11.71+17.57)/2=14.6
- Heading change .5 sec * 14.6^{o}/sec = 7.3 deg
- Total heading change 3.4+7.3 = 10.7 degrees
- Change in CG approach angle less than that?

TIME DISTANCE INTEGRATION

- Copy speed in mph vs time data into Microsoft Excel
- Create new column with speed in Feet per Second
- Create new column with FPS * time interval, the distance in feet moved during that time interval
- Add the distances moved in the next to last

TIME	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0
SPEED MPH	50.3	50.3	48.5	45.4	41.6	36.7	29.8	22.4	13	7.5	8.1
SPEED FPS	73.7	73.7	71.1	66.6	61.0	53.8	43.7	32.8	19.1	11.0	11.9
DIST IN 0.5 SEC	36.9	36.9	35.6	33.3	30.5	26.9	21.8	16.4	9.5	5.5	5.9
CUMULATIVE dist	259.2	222.3	185.4	149.9	116.6	86.1	59.2	37.4	21.0	11.4	5.9


Speed vs Dist to Impact
60
50
40
30
20
10
0
Speed in MPH
Series1
Initial Light Braking @ -4
Moderate Braking @ -3
Panic Braking @ -2
300
250
200
150
100
50
0
Distance to Second Impact in Feet (First Impact 10 ft less)

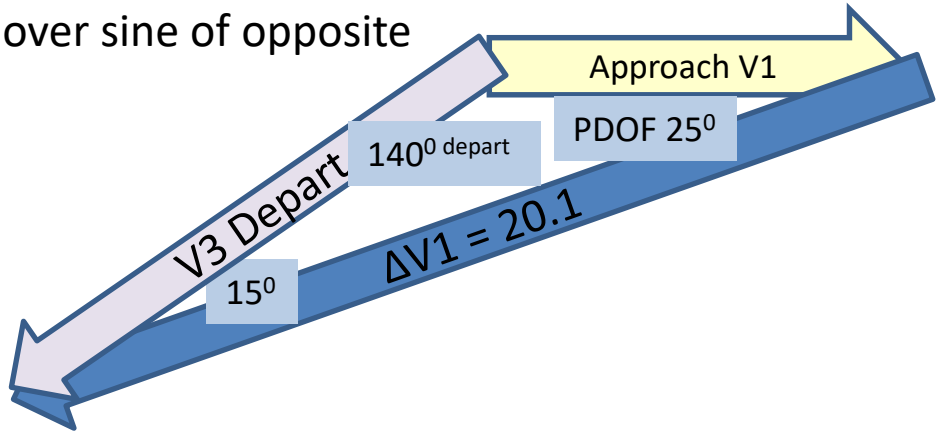
Toyota Velocity Vectors "Triangle"

Toyota Knowns
Weight = 3500
DeltaV = 20.1 mph
PDOF = +155 degrees (-25 +180)
Approach = 0 degrees (by choice)
Departure = 220 Degrees from 0 (140 in triangle)

Set Up your Triangle – Need 3 Pieces of Info to Solve Triangle
You have PDOF Angle, Delta V and Departure Angle !!
Two angles and one side – that's enough!!
Calculate 3rd angle using 180-140-25 = 15 degrees
Is it a right triangle??? NO – can't use regular Trig
Do you have any opposite pairs? YES – Use LAW OF SINES
Set up your Law of Sines equation – side over sine of opposite angles are equal for all 3 sides/angles

$$\frac{20.1}{Sin(140)} = \frac{V1}{Sin(15)}$$

$$\frac{20.1}{Sin(140)} = \frac{V3}{Sin(25)}$$



Toyota Vectors

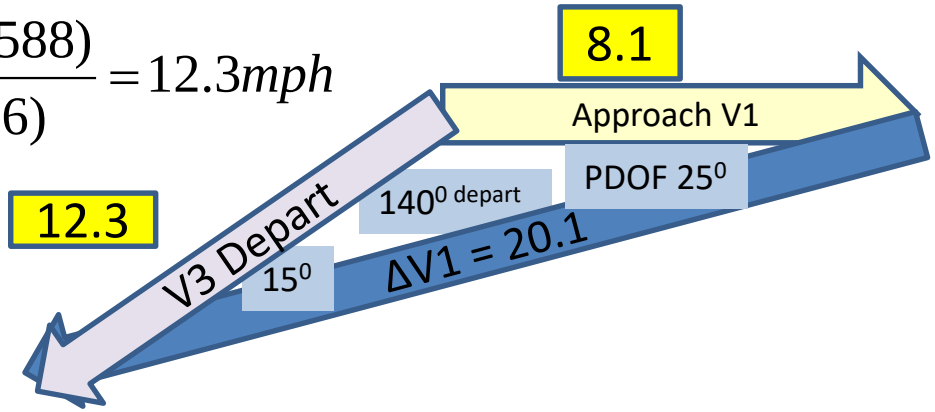
Toyota Knowns
Weight = 3500
DeltaV = 18.2 mph
PDOF = 25 degrees
Approach = 0 degrees
Departure = 220 Degrees

NOW FILL IN THE KNOWNS INTO YOUR LAW OF SINES EQUATION AND SOLVE FOR V1 and V3

$$\frac{20.1}{Sin(140)} = \frac{V1}{Sin(15)} \qquad V1 = \frac{20.1(.2588)}{(.6428)} = 8.1mph$$

$$\frac{20.1}{Sin(140)} = \frac{V3}{Sin(25)} \qquad V3 = \frac{20.1(.2588)}{(.4226)} = 12.3mph$$

Compare with 8.1 mph from Pre-crash Data

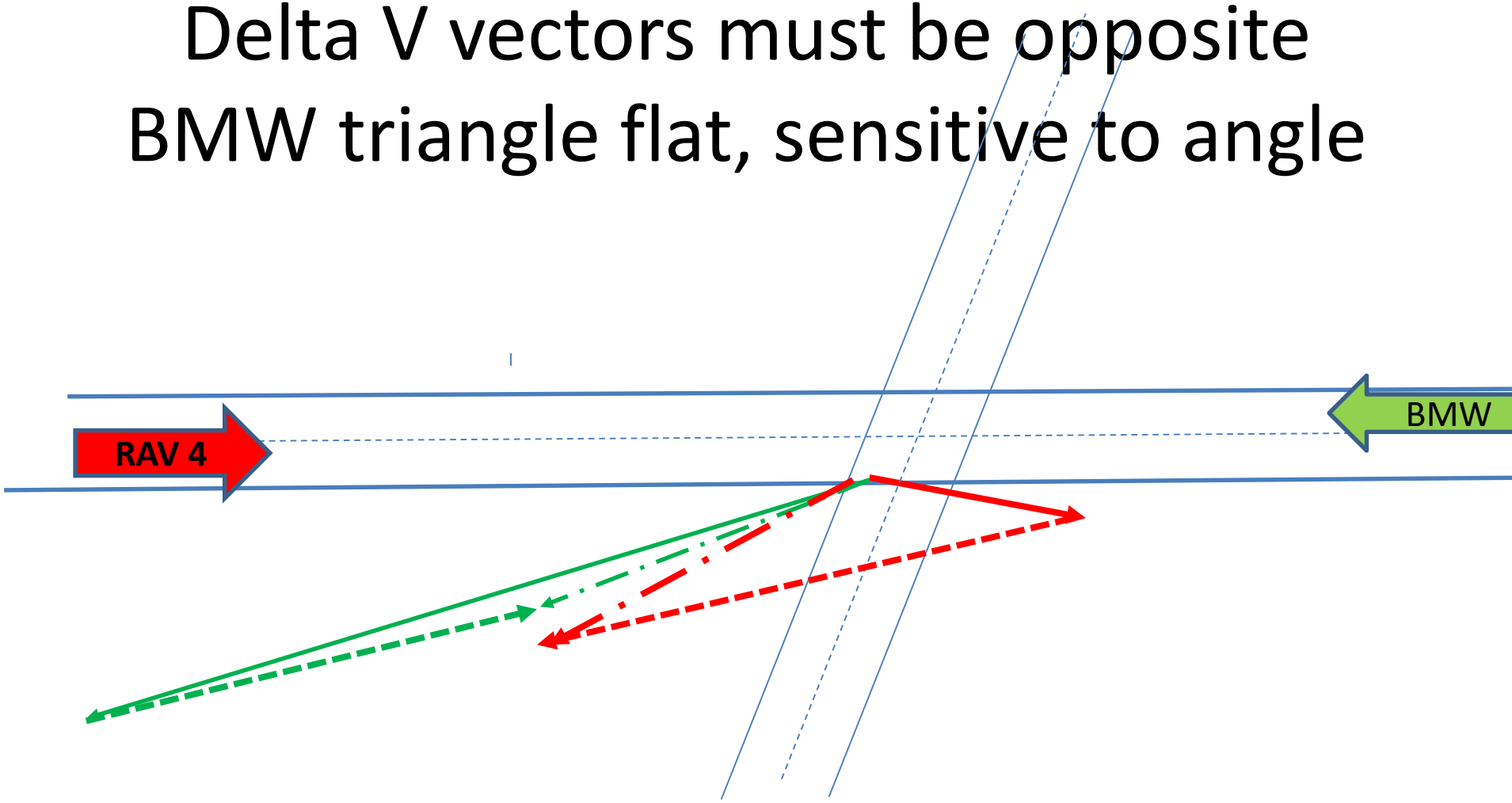


What do we know about the BMW?

BMW Knowns
Weight = approx. 3500 lbs
DeltaV = approx. 20.1 mph

Forces in a crash are equal and opposite
Delta V Magnitudes are inversely proportional to Weight
Weights are about equal – BMW Delta V magnitude approximately 20
IF the vehicles reached a common velocity, Closing speed is approximately sum of two Delta V's (adjust for restitution if present)
Sum of Delta v's is about 40 mph
155
RAV4 was going about 8 mph
Technically need difference in approach angles, but for first pass assume the BMW speed at impact is 40-8 = 32 mph (once you figure the angles do the real math)

RAV 4 vs BMW 3 series
Draw approximate velocity vectors
Delta V vectors must be opposite
BMW triangle flat, sensitive to angle



Final Conclusions

- RAV4 clearly did not steer left into BMW's lane before it swerved right just prior to impact based on EDR steering data
- RAV4 showed initial reaction to BMW getting out of its lane at -4 seconds, and increasing braking rate as V2 got closer – not an immediate panic reaction that would be expected if V1 driver "woke up". RAV4 saw BMW weaving, then BMW tries to turn left?
- RAV4 speed at impact is low, 8 mph, almost stopped. But BMW is still moving at 32 mph. Is BMW making sweeping turn at speed?

Toyota EDR Summary

- Toyota began installing EDR's in 2001MY; many Gen 1 and some Gen 2 have Delta V data only, others have DV and precrash data. Gen 1 & 2 data is similar to GM but has added precrash data point sampled at TRG.
- Unique Gen 1 &2 issues (78 mph max speed, accel pedal in volts, & last data point up to 0.5 seconds old) require special attention.
- Toyota EDR data is as ***reliable*** as other manufacturers within the data limitations.
- Newer "12EDR" even better plus many additional data elements helpful to crash recon. "13EDR" better yet!
- New Vehicle Control History gives driving habits